

NEW 2026 Curriculum

Free Chapter: Year 11 Extension 1 Maths

Polynomials

- ✓ Fully worked solutions
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- ✓ Theory Explanations

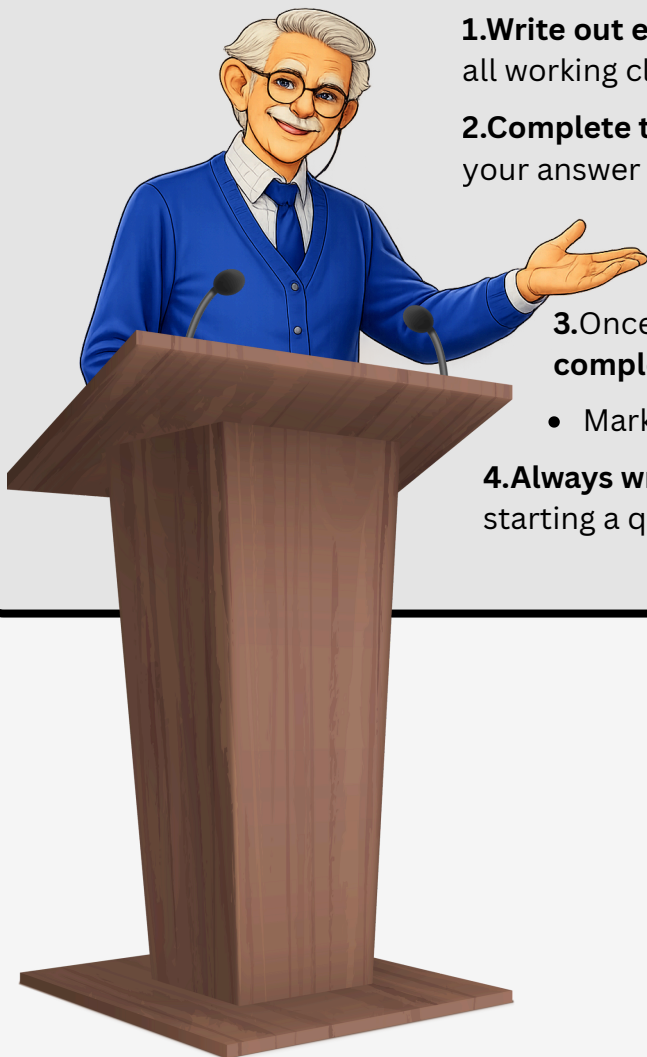
**John Kinny-Lewis**

A Proven Way to Learn Mathematics

With over **45 years of classroom teaching experience**, John Kinny-Lewis has developed a proven approach to learning mathematics that builds confidence, accuracy, and understanding.

This book is designed to be used actively – not just read.

Follow these steps for best results:



1. Write out each question in your workbook and complete all working clearly.

2. Complete the odd-numbered questions first, checking your answer and the worked solution after each question.

- Use the worked solutions to identify and correct any mistakes immediately.

3. Once the odd questions are complete, return and complete the even-numbered questions.

- Mark your work as you go and fix any errors.

4. Always write down the formula you are using before starting a question (for example, Area of a rectangle).

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Polynomials

1

Remainder Theorem

Theory

If a polynomial $P(x)$ is divided by $(x-a)$ until the remainder R does not contain x , then $R = P(a)$.

For example: $P(x) = (x-2)Q(x) + R$, $P(2) = (2-2)Q(x) + R \rightarrow R = P(2)$

Note that $P(x) = (x+2)Q(x) + R$, $P(-2) = (-2+2)Q(x) + R \rightarrow R = P(-2)$

Example 1

Find the remainder when $P(x) = 3x^2 + 7x - 4$ is divided by $x - 4$.

Example 1 solution

$$P(x) = 3x^2 + 7x - 4$$

$$P(4) = 3(4)^2 + 7(4) - 4$$

$$= 48 + 28 - 4$$

$$= 72$$

$$\therefore 3x^2 + 7x - 4 = (x - 4)Q(x) + 72$$

$$\therefore \text{Remainder} = 72$$

Example 2

If $P(x) = x^4 - x^3 - ax^2 - 4x + b$, find a and b if $P(0) = 2$ and $P(1) = 5$.

Example 2 solution

$$P(0) = b$$

$$P(0) = 2$$

$$\therefore b = 2$$

$$\therefore P(1) = 1 - 1 - a - 4 + b$$

$$= -a + b - 4$$

$$= -a + 2 - 4$$

$$= -a - 2$$

$$P(1) = 5$$

$$\therefore -a - 2 = 5$$

$$a = -7$$

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1. When $P(x) = ax^3 - 5x + 8$ is divided by $(x - 1)$ the remainder is 2. The value of a is?
2. When $P(x) = x^3 - 2x^2 - ax + 6$ is divided by $(x + 1)$ the remainder is 7. The value of a is?
3. The polynomial $P(x) = x^2 + ax + b$ has a zero at $x = -3$. When $P(x)$ is divided by $x - 4$, the remainder is 21. The values of a and b are?
4. $P(x) = x^4 - 3x^3 + ax^2 + bx - 6$ leaves a remainder of 8 when divided by $x + 1$. Given that $x - 3$ is a factor of $P(x)$ then the values of a and b are?
5. Suppose $x^3 - 2x^2 + a = (x + 2)Q(x) + 3$ where $Q(x)$ is a polynomial. Find the value of a .
6. When $P(x) = ax^3 + bx^2 + cx + d$ is divided by $(x - \alpha)$ the remainder is p , and when divided by $(x + \alpha)$ the remainder is $-p$. Prove that: $\alpha = \frac{pb}{bc - ad}$
7. The polynomial $P(x) = (x - p)^3 + q$ is zero at $x = 1$ and when divided by x , the remainder is -7. Find all possible values of the pair p, q .
8. Use the remainder theorem to find one factor of $P(x) = x(x + m) - n(n + m)$. By division, or otherwise, find the other factor.
9. When the polynomial $P(x) = 2x^3 - 3mx^2 - 7x - 2$ is divided by $x - 2$, the remainder is 12. Find the value of m .
10. Find the quotient and remainder when $P(x) = -2x^3 + 7x^2 - 7x + 1$ is divided by $2x + 3$.
11. The polynomial $P(x) = 2x^3 + ax^2 + bx + 6$ is divisible by $x - 1$ and leaves the remainder of -12 when divided by $x + 2$. The values of a and b are?
12. Find the quotient and remainder when $P(x) = x^3 - x^2 + x - 1$ is divided by $x - 1$
13. Find the quotient and remainder when $P(x) = x^4 + 3x^3 + 5x^2 - 6x - 8$ is divided by $x + 2$
14. Divide the polynomial $P(x) = 2x^4 - 10x^3 + 12x^2 + 2x - 3$ by $x^2 - 3x + 1$. Hence determine the quotient and the remainder.
15. Find the quotient and remainder when $P(x) = x^3 - 7x - 3$ is divided by $x - 2$.
16. Find the quotient and remainder when $P(x) = x^4 - 3x^3 - 4$ is divided by $x^2 + 2$
17. Find the remainder when $P(x) = x^4 - x^2 + 1$ is divided by $x^2 + 1$
18. When $P(x) = x^3 + 3x^2 + 7x - b$ is divided by $x - 2$ the remainder is 20. Find the value of b .
19. Find the remainder when $P(x) = x^3 + 3x^2 + 2x - 6$ is divided by $x - 2$
20. When $P(x) = 2x^3 + 7x^2 + ax + b$ is divided by $x - 3$ the remainder is 120 and when $P(x)$ is divided by $x + 1$ the remainder is -8. Find a and b .
21. When $P(x) = x^4 + 5x^3 - ax + b$ and $Q(x) = ax^2 + bx - 1$ are each divided by $x + 1$, the remainders are 7 and -6 respectively. Find a and b .
22. When $P(x) = ax^2 + bx + c$ is divided by $x - p$ the remainder is q .
When $P(x)$ is divided by $x + p$ the remainder is r .
Prove that $p = \frac{q - r}{2b}$
23. When $P(x) = 3x^3 - ax^2 - bx + 1$ is divided by $x - 2$ the remainder is 15. When $P(x)$ is divided by $x - 1$ there is no remainder. Find a and b .
24. When $P(x) = x^3 + 3x^2 - mx + n$ is divided by $x + 2$ the remainder is 9. When $P(x)$ is divided by $x - 3$ the remainder is 49. Find m and n .

- 25.** If $P(x) = x^3 + 2x^2 + 3x + q$ and $Q(x) = x^3 + x^2 + 9$ leave the same remainder when divided by $x + 2$. Find q .
- 26.** When the polynomial $P(x) = x^4 - 5x^3 + 6x^2 - ax + b$ is divided by $x - 3$, the remainder is 20 . When $P(x)$ is divided by $x + 2$ the remainder is 30. Find the values of a and b .
- 27.** Suppose $x^3 - 4x^2 + k \equiv (x - 2)Q(x) + 4$. Find the value of k .
- 28.** When a polynomial is divided by $x^2 - 9$ the remainder is $x + 1$. What is the remainder when the polynomial is divided by $x - 3$?
- 29.** $P(x) = x^3 - kx^2 + 2x$, where k is a real number. When $P(x)$ is divided by $x - 2$ the remainder is 4. Find the remainder when $P(x)$ is divided by $x - 1$.
- 30.** The polynomial $P(x) = x^3 - ax + b$ has a remainder of 4 when divided by $x - 2$ and a remainder of 8 when divided by $x + 2$. Find the values of a and b .
- 31.** Let $P(x) = (x - 1)(x + 2)Q(x) + a(x - 1) + b$, where $Q(x)$ is a polynomial and a and b are real numbers.
 When $P(x)$ is divided by $(x - 1)$ the remainder is 10.
 When $P(x)$ is divided by $(x + 2)$ the remainder is -2 .
- i) What is the value of b ?
- ii) What is the remainder when $P(x)$ is divided by $(x - 1)(x + 2)$?

Past HSC Questions

- 32.** HSC 2014 - Question 9
- 33.** HSC 2015 - Question 1
- 34.** HSC 2016 - Question 2
- 35.** HSC 2019 - Question 11d

- 36.** HSC 2021 - Question 3
- 37.** HSC 2022 - Question 3
- 38.** HSC 2023 - Question 11c

Theory

For a polynomial $P(x)$, if $P(a) = 0$ then $(x - a)$ is a factor of $P(x)$.

Conversely if $(x - a)$ is a factor of $P(x)$ then $P(a) = 0$.

For example: $P(x) = (x - 2)Q(x)$, $(x - 2)$ is a factor of $P(x)$. $P(2) = (2 - 2)Q(2) \rightarrow P(2) = 0$

Example

Let $P(x) = 6x^3 - 11x^2 - 26x - 40$

i) What of the following is a factor of $P(x)$

- a) $(x - 8)$
- b) $(x - 5)$
- c) $(x - 10)$
- d) $(x - 2)$

ii) Write the remaining factor obtained after dividing $P(x)$ by the factor in (i)

Solution

$$P(x) = 6x^3 + 11x^2 - 26x - 40$$

(i) We need to check for the factors of $P(x)$

If $(x - a)$ is a factor of $P(x)$ then $P(a) = 0$

$$(A) P(8) = 3072 + 704 - 208 - 40 = 3528 \neq 0$$

$$(B) P(5) = 750 + 275 - 130 - 40 = 855 \neq 0$$

$$(C) P(10) = 6000 + 1100 - 260 - 40 = 7400 \neq 0$$

$$(D) P(2) = 48 + 44 - 52 - 40 = 0$$

So $(x - 2)$ is a factor of $P(x)$

To find other factor of $P(x)$, we follow long division

$$\begin{array}{r}
 6x^2 + 23x + 20 \\
 x - 2 \overline{) 6x^3 + 11x^2 - 26x - 40} \\
 \underline{- 6x^3 + 12x^2} \\
 23x^2 - 26x \\
 \underline{- 23x^2 + 46x} \\
 20x - 40 \\
 \underline{- 20x + 40} \\
 0
 \end{array}$$

Hence the remaining factor is $(6x^2 + 23x + 20)$

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39. Find the value of k if $x + 3$ is a factor of $P(x) = x^3 - 2kx - 3$.
40. $x^3 + ax + b$ is divisible by $(x - 3)$ and $(x - 1)$. The values of a and b are?
41. Given that $(x - 3)$ is a factor of $x^3 - 13x + a$ then the value of a is?
42. For what value of b is $(x + 1)$ a factor of $x^3 - bx + 3$?
43. The value of k for which $(x + 2)$ is a factor of $2x^3 + kx^2 + 5x - 2$ is?
44. Find the values of a and b if $(x + 1)(x - 1)$ is a factor of $x^3 + ax^2 + bx - 4$.
45. Use the factor theorem to fully factorize: $P(x) = 3x^3 + 10x^2 + 9x + 2$
46. Find the values of a and b if $(x - 1)(x + 1)$ is a factor of $P(x) = x^3 + ax^2 + bx - 4$ and find the other factor.
47. Find the values of a and b such that $P(x) = x^4 + x^3 + x^2 + ax + b$ is divisible by $x^2 - 3x + 2$.
48. What are the roots of $x^3 + 5x^2 + 8x + 4 = 0$
49. $P(x) = x^3 + 3x^2 - 10x - 24$, find $P(-2)$ and express $P(x)$ as the product of three linear factors.
50. Solve $P(x) = x^3 - 3x^2 - 8x + 4$ over the real number field.
51. Find the quotient and remainder when $P(x) = x^4 + 3x^3 + 5x^2 - 6x - 8$ is divided by $x + 2$.
52. Divide the polynomial $P(x) = 2x^4 - 10x^3 + 12x^2 + 2x - 3$ by $x^2 - 3x + 1$. Hence determine the quotient and the remainder.
53. Find the quotient and remainder when $P(x) = x^3 - 7x - 3$ is divided by $x - 2$.
54. Find the quotient and remainder when $P(x) = x^4 - 3x^3 - 4$ is divided by $x^2 + 2$
55. Find the remainder when $P(x) = x^4 - x^2 + 1$ is divided by $x^2 + 1$
56. Find the quotient and remainder when $P(x) = -2x^3 + 7x^2 - 7x + 1$ is divided by $2x + 3$
57. $P(x) = x^3 - x^2 - 10x - 8$. Use the factor theorem to fully factorise $P(x)$.
58. Show that $P(x) = x^3 + 8$ has only one linear factor over the real number field.
59. Show that $P(x) = x^3 + 8$ has only one linear factor over the real number field.
60. When $P(x) = 3x^3 - ax^2 - bx + 1$ is divided by $x - 2$ the remainder is 15. When $P(x)$ is divided by $x - 1$ there is no remainder. Find a and b .
61. When $P(x) = x^3 + 3x^2 - mx + n$ is divided by $x + 2$ the remainder is 9. When $P(x)$ is divided by $x - 3$ the remainder is 49. Find m and n .
62. If $P(x) = x^3 + 2x^2 + 3x + q$ and $Q(x) = x^3 + x^2 + 9$ leave the same remainder when divided by $x + 2$. Find q .
63. When the polynomial $P(x) = x^4 - 5x^3 + 6x^2 - ax + b$ is divided by $x - 3$, the remainder is 20. When $P(x)$ is divided by $x + 2$ the remainder is 30. Find the values of a and b .
64. Suppose $x^3 - 4x^2 + k \equiv (x - 2)Q(x) + 4$. Find the value of k
65. When a polynomial is divided by $x^2 - 9$ the remainder is $x + 1$. What is the remainder when the polynomial is divided by $x - 3$?
66. The polynomial $P(x) = x^3 - ax + b$ has a remainder of 4 when divided by $x - 2$ and a remainder of 8 when divided by $x + 2$. Find the values of a and b .
67. Let $P(x) = (x - 1)(x + 2)Q(x) + a(x - 1) + b$, where $Q(x)$ is a polynomial and a and b are real numbers.

When $P(x)$ is divided by $(x - 1)$ the remainder is 10 .

When $P(x)$ is divided by $(x + 2)$ the remainder is -2 .

i) What is the value of b ?

ii) What is the remainder when $P(x)$ is divided by $(x - 1)(x + 2)$?

Past HSC Questions

68. HSC 2015 - Question 11f

69. HSC 2017 - Question 1

70. HSC 2018 - Question 11a

71. HSC 2019 - Question 7

72. HSC 2020 - Question 11a

Theory

Quadratic equations

For the quadratic equation $ax^2 + bx + c = 0$.

The sum of the roots $\alpha + \beta = -\frac{b}{a}$

The product of the roots $\alpha\beta = \frac{c}{a}$

Cubic equations

For the cubic equation $ax^3 + bx^2 + cx + d = 0$.

The sum of the roots $\sum \alpha = \alpha + \beta + \gamma = -\frac{b}{a}$.

The sum of products of each pair of roots $\sum \alpha\beta = \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$

The product of the roots $\alpha\beta\gamma = -\frac{d}{a}$

Quartic Equations

For the quartic equation $ax^4 + bx^3 + cx^2 + dx + e = 0$

The sum of the roots $\sum \alpha = \alpha + \beta + \gamma + \delta = -\frac{b}{a}$

The sum of products of each pair of roots $\sum \alpha\beta = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{c}{a}$

The sum of products of each triplet of roots $\sum \alpha\beta\gamma = \alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -\frac{d}{a}$

The product of the roots $\alpha\beta\gamma\delta = \frac{e}{a}$

Example

The polynomial $P(x) = x^3 - x^2 + 4x - 6$ has roots α, β and γ . Evaluate:

- i) $\alpha + \beta + \gamma$
- ii) $\alpha\beta\gamma$
- iii) $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma}$

Solution

i) $\alpha + \beta + \gamma = 1$

ii) $\alpha\beta\gamma = 6$

iii) $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} = \frac{\gamma + \alpha + \beta}{\alpha\beta\gamma}$
 $= \frac{1}{6}$

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73. The cubic polynomial $P(x) = x^3 + ax^2 + bx + c$, where a , b and c are real numbers has three real zeros, 1, α and $\frac{1}{\alpha}$. The value of $a + b$ is?
74. It is known that two of the roots of the equation: $2x^3 - 2x^2 - kx + 6 = 0$ are equal in value but opposite in sign. Find the value of k .
75. $P(x) = x^3 - 6x^2 + 11x - 6$ has one of the roots equal to the sum of the other two roots. The values of the three roots are?
76. If α and β are the roots of the equation $x^2 + 4x - 2 = 0$, then the quadratic equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is?
77. The roots of $x^3 - 6x^2 + 3x + m = 0$ are in arithmetic progression. The value of m is?
78. The product of two roots of the polynomial equation $2x^3 - kx^2 - 8x + 4 = 0$ is 1. Find the value of k .
79. Give that the equation $x^3 - 7x^2 + 16x + k = 0$ has two equal and integral roots, find the value of k .
80. If α and β are the roots of the equation $x^2 + 8x - 5 = 0$, find the quadratic equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.
81. One of the roots of $P(x) = x^3 + px^2 + 1$ is equal to the sum of the other two roots of $P(x)$. Find the value of p .
82. The polynomial $P(x)$ is given by $P(x) = px^3 - 8x^2 + qx - 32$, where p and q are constants. The three zeros of $P(x)$ are $-2, 4$ and α . Find the value of α .
83. Solve $4x^3 + 32x^2 + 79x + 60 = 0$ given that one root is equal to the sum of the other two roots.
84. Two of the roots of $P(x) = 2x^3 + 7x^2 + 7x + 2$ are reciprocals of each other. Find the roots of the equation.
85. Solve $2x^3 - 21x^2 + 42x - 16 = 0$ given that the roots form consecutive terms of a geometric progression.
(Let the roots be a, ar and ar^2)
86. If α, β and γ are the roots of $3x^3 + 8x^2 - 1 = 0$, find the value of $\left(\beta + \frac{1}{\gamma}\right)\left(\gamma + \frac{1}{\alpha}\right)\left(\alpha + \frac{1}{\beta}\right)$.
87. Solve $x^4 - 2x^3 - 3x^2 + 4x + 2 = 0$, given that the sum of two of its roots are zero.
88. Solve the equation $8x^3 - 36x^2 + 22x + 21 = 0$, given that the roots are in arithmetic progression.
Let the roots be $a - d, a$ and $a + d$.
89. If α, β, γ are the roots of $P(x) = x^3 - x^2 + 4x - 1$ find the value of $(\alpha + 1)(\beta + 1)(\gamma + 1)$.
90. One of the roots of $P(x) = x^3 - 8x + 19x - 12$ is equal to the sum of the other two roots. Find the roots of the equation.
91. It is known that two roots of $P(x) = 2x^3 + x^2 - kx + 4 = 0$ are reciprocals of each other. Find the value of k .
92. Solve the equation $x^4 + 4x^3 - x^2 - 16x - 12 = 0$ given that the sum of two of the roots is zero.
93. Solve the equation $x^4 - 5x^3 + 4x^2 + 10x - 12 = 0$ given that the product of two of the roots is 6.

- 94.** If α and β are the roots of $x^2 + mx + n = 0$, find the roots of $nx^2 + (2n - m)x + n = 0$ in terms of α and β
- 95.** The polynomial $P(x) = x^3 + ax^2 + bx + c$, where a, b and c are real numbers, has three real zeros: $-1, \alpha, -\alpha$.
- Find the value of a .
 - Find the value of $c - b$.
- 96.** The polynomial $P(x)$ is given by $P(x) = px^3 + 8x^2 + qx - 32$, where p and q are constants. The three zeros of $P(x)$ are $-2, 4$ and α . Find the value of α .
- 97.** The polynomial $P(x) = 2x^4 - 4x^2 - 6x + 3$ has roots $\alpha, \beta, \gamma, \delta$. Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$.
- 98.** Solve the equation $x^3 - 5x^2 + 3x + 9 = 0$ given that two of the roots are equal.
- 99.** If α and β are the roots of the equation $2x^2 - 4x + 3 = 0$ find:
- $\frac{1}{\alpha} + \frac{1}{\beta}$
 - $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$
- 100.** The polynomial $P(x) = x^3 - 4x^2 + kx + 16$ has roots α, β, γ .
- Find the value of $\alpha + \beta + \gamma$.
 - Find the value of $\alpha\beta\gamma$.
 - It is known that two of the roots are equal in magnitude but opposite in sign. Find the third root and the value of k .
 - Find the other two roots.
- 101.** If the roots of $x^2 + px + q = 0$ are α and β express in terms of p and q .
- $\frac{1}{\alpha} + \frac{1}{\beta}$
 - $\alpha^2 + \beta^2$
 - $(\alpha + 1)(\beta + 1)$
- 102.** α, β, γ are the roots of $P(x) = x^3 - 2x + 1$, find:
- $\alpha + \beta + \gamma$
 - $\alpha\beta\gamma$
 - $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

103. *NOTE: This is a very challenging and hard question.

Express $\cos 3\theta = -\frac{1}{2}$ in terms of $\cos \theta$ and by letting $x = \cos \theta$ show that $\cos 3\theta = -\frac{1}{2}$ can be expressed as $8x^3 - 6x + 1 = 0$. Hence deduce that:

i) $\cos \frac{\pi}{9} = \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9}$

ii) $\sec \frac{\pi}{9} \sec \frac{2\pi}{9} \sec \frac{4\pi}{9} = 8$

iii) $\tan^2 \frac{\pi}{9} + \tan^2 \frac{2\pi}{9} + \tan^2 \frac{4\pi}{9} = 33$

104. *Note: This is a very hard and challenging question.

i) Prove that $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$

ii) Let $x = \tan \theta$, and $\tan \theta = 1$ hence find the three roots of $x^3 - 3x^2 - 3x + 1 = 0$

iii) Find the value of
(α) $\tan \frac{\pi}{12} + \tan \frac{5\pi}{12}$
(β) $\tan \frac{\pi}{12} \times \tan \frac{5\pi}{12}$

Past HSC Questions

105. HSC 2014 - Question 5

106. HSC 2018 - Question 1

107. HSC 2021 - Question 11h

108. HSC 2022 - Question 13d

Theory

If $x = b$ is a root of multiplicity r of a real polynomial equation $P(x) = 0$, then $x = b$ is also a root of the derived polynomial equation $\frac{dP}{dx} = 0$ of multiplicity $(r - 1)$.

Consider $P(x) = (x - b)^r \times Q(x)$ where $Q(b) \neq 0$.

$$\frac{dP}{dx} = r(x - b)^{r-1} \times Q(x) + \frac{dQ}{dx} \times (x - b)^r$$

$$\frac{dP}{dx} = (x - b)^{r-1} \left\{ rQ(x) + (x - b) \frac{dQ}{dx} \right\}$$

$\therefore x = b$ is a root of multiplicity $(r - 1)$.

Example

Solve the equation $2x^3 - 15x^2 + 24x + 16 = 0$ given that it has a double root.

Solution

$$P(x) = 2x^3 - 15x^2 + 24x + 16$$

$$P'(x) = 6x^2 - 30x + 24$$

$$= 6(x^2 - 5x + 4)$$

$$= 6(x - 4)(x - 1)$$

$$P(1) = 2 - 15 + 24 + 16 \neq 0$$

$$P(4) = 128 - 240 + 96 + 16 = 0$$

$\therefore x = 4$ is a double root of $P(x)$

$$2x^3 - 15x^2 + 24x + 16 \equiv (x - 4)^2(2x - c)$$

$$\text{Let } x = 0 \rightarrow 16 = 16 \times -c$$

$$\therefore c = -1$$

$$P(x) = (x - 4)^2(2x + 1)$$

$$\therefore x = 4, 4 \text{ and } -\frac{1}{2}$$

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- 109.** $2x^3 - 15x^2 + 24x + 16 = 0$, has a root of multiplicity of 2. The root is?
- 110.** $4x^3 + 12x^2 - 15x + 4 = 0$ has a root of multiplicity of 2. The root is?
- 111.** $x^4 + 2x^3 - 12x^2 + 14x - 5 = 0$ has a root of multiplicity of 3. The root is?
- 112.** $P(x) = x^3 - 6x^2 + 9x + k$ has a root of multiplicity of 2. A possible value of k is?
- 113.** $P(x) = x^4 + ax^3 + bx^2 + cx - 16$ has a triple root at $x = 2$. The values of a , b and c are?
- 114.** Solve the equation $x^4 + 2x^3 - 12x^2 + 14x - 5 = 0$ given that it has a triple root.
- 115.** Find the values of p and q given that $P(x) = 2x^3 - (2p + 1)x^2 + (2 + q)x - 1$ has double root at $x = 1$.
- 116.** $P(x) = x^4 - 3x^3 - 6x^2 + 28x - 24$ has a triple zero, find all the zeros of $P(x)$.
- 117.** $P(x) = x^3 + x^2 - 5x + 3$ has a root of multiplicity of 2. The roots are
- 118.** Given that $P(x) = x^4 + 4x^3 - 16x + k$ has a root of multiplicity of 3 and $k < 0$. Find the value of k .
- 119.** The polynomial $P(x) = x^3 - 6x^2 + 9x + k$ has a double root. Find the two possible values for k and all the roots for each value of k .
- 120.** The polynomial $P(x) = x^3 + 3x^2 - 24x + k$ has a double root. Find the two possible values of k and all the roots.
- 121.** Find the real roots of $P(x) = x^3 - 7x^2 + 11x - 5$ given that $P(x)$ has a double root.
- 122.** The line $y = 4x - 5$ is a tangent to the curve $y = x^3 - 2x^2 + 3$ at the point $(2, 3)$. The tangent meets the curve again at Q . Find the coordinates of Q .
- 123.** If $P(x) = x^3 - 3x^2 + 4$, show that $P(x)$ has a multiple zero and find the zero and its multiplicity.
- 124.** If $px^3 + qx^2 + r = 0$ has a double root, show that $27p^2r + 4q^3 = 0$.

Past HSC Questions

- 125.** HSC 2023 - Question 14b

Theory

For a polynomial of odd degree always has at least one real zero. The graph must cut the x -axis at least once.

At least one maximum or minimum value of $y = P(x)$ occurs between any two distinct zeros.

For a polynomial of odd degree, the ends of the graph go in opposite directions.

For a polynomial of even degree, the ends of the graph go in the same direction.

When the graph of $y = P(x)$ meets the x -axis, it may cut (single zero), touch (double zero) or cut it at a point of inflexion (triple zero).

Example

Find the roots of the polynomial

$P(x) = x^4 + 3x^3 - 3x^2 - 7x + 6$ given that $x = 1$ is a double root. Hence sketch $y = P(x)$.

Solution

$$(x - 1)^2 = x^2 - 2x + 1$$

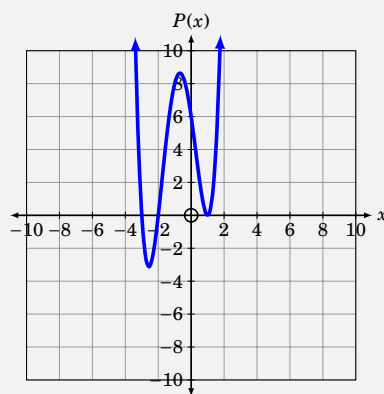
$$\begin{array}{r}
 x^2 + 5x + 6 \\
 x^2 - 2x + 1 \sqrt{x^4 + 3x^3 - 3x^2 - 7x + 6} \\
 \underline{x^4 - 2x^3 + x^2} \\
 5x^3 - 4x^2 - 7x + 6 \\
 \underline{5x^3 - 10x^2 + 5x} \\
 6x^2 - 12x + 6 \\
 \underline{6x^2 - 12x + 6} \\
 0
 \end{array}$$

$$\therefore P(x) = (x - 1)^2(x^2 + 5x + 6)$$

$$= (x - 1)^2(x + 2)(x + 3)$$

$$\therefore \text{Roots are } x = 1, x = -2, x = -3$$

y intercept is 6



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- 126.** Sketch the graph of $y = (x - 1)^3 + 2$
- 127.** Sketch the graph of $y = 3 - 2(x + 1)^3$
- 128.** Sketch the graph of $y = (x + 1)^2(2 - x)^2$
- 129.** Sketch the graph of $y = (x + 1)(2 - x)^3$
- 130.** Sketch the graph of $y = (x + 1)^3(2 - x)^2$
- 131.** Sketch the graph of $y = x(x - 2)^2$
- 132.** Sketch the graph of $y = (x + 1)^3(2 - x)$ showing all x and y intercepts and point(s) of inflexion.
- 133.** Sketch the graph of $y = (x + 1)(x - 2)(x - 1)$ showing all x and y intercepts.
- 134.** Sketch the graph of $y = x(x + 2)(x - 1)^2$ showing all x and y intercepts and point(s) of contact.
- 135.** Sketch the graph of $y = (x - 1)^2(x + 2)^2$
- 136.** Sketch the graph of $y = (x + 1)(x - 2)(x - 1)$ showing all x and y intercepts.
- 137.** Factorise $P(x) = x^2 - x^4$, find the zeros of $P(x)$ and sketch the polynomial.
- 138.** Given that $x = 2$ is the zero of the polynomial $P(x) = x^3 - 3x - 2$, find the other zeros, sketch the polynomial and hence solve the inequality $x^3 - 3x - 2 > 0$.
- 139.** Find the roots of the polynomial $P(x) = x^4 + 3x^3 - 3x^2 - 7x + 6$ given that $x = 1$ is a double root. Hence sketch $y = P(x)$.
- 140.** Find the zeros of the polynomial $P(x) = 5x(x - 1)^3$ and sketch the polynomial.
- 141.** Factorise $P(x) = x^3 - x^2 - x + 1$, find the zeros of $P(x)$ and sketch the polynomial.
- 142.** Find the zeros of the polynomial $P(x) = \frac{1}{3}x^3(x - 2)^2(x + 2)$ and sketch the polynomial.
- 143.** Find the zeros of the polynomial $P(x) = x(x + 2)(x - 1)^3$ and sketch $y = P(x)$
- 144.** Find the zeros of the polynomial $P(x) = -x^4 + 8x^2 - 16$ and sketch the polynomial.
- 145.** The polynomial $P(x) = x^3 + px^2 + qx + r$ has roots $0, 2, -2$.
- Find p, q and r .
 - Without calculus sketch the graph of $y = P(x)$.
 - Hence, solve the inequality $x^3 - 4x \geq 0$.

Past HSC Questions

- 146.** HSC 2016 - Question 10
- 147.** HSC 2018 - Question 4

- 148.** HSC 2020 - Question 5

Worked Solutions

Remainder Theorem

1. $P(x) = ax^3 - 5x + 8$

$$P(1) = a(1)^3 - 5 \times 1 + 8 = 2$$

$$a + 3 = 2$$

$$\therefore a = -1$$

2. $P(-1) = (-1)^3 - 2(-1)^2 - a \times -1 + 6$

$$7 = -1 - 2 + a + 6$$

$$7 = a + 3$$

$$\therefore 4 = a$$

3. $P(x) = x^2 + ax + b$

$$P(-3) = (-3)^2 + a(-3) + b = 0$$

$$-3a + b = -9 \dots (1)$$

$$P(4) = (4)^2 + a(4) + b = 21$$

$$4a + b = 5 \dots (2)$$

$$(1) - (2) \quad -7a = -14 \rightarrow a = 2$$

$$\text{In (2)} \quad 4 \times 2 + b = 5 \rightarrow b = -3$$

4. $P(x) = x^4 - 3x^3 + ax^2 + bx - 6$

$$P(-1) = (-1)^4 - 3 \times (-1)^3 + a \times (-1)^2 - b - 6 = 8$$

$$1 + 3 + a - b - 6 = 8$$

$$a - b = 10 \dots (1)$$

$$P(3) = 3^4 - 3 \times 3^3 + a \times 3^2 + 3b - 6 = 0$$

$$81 - 81 + 9a + 3b - 6 = 0$$

$$9a + 3b = 6$$

$$3a + b = 2 \dots (2)$$

$$(1) + (2) \quad 4a = 12 \rightarrow a = 3$$

$$\text{In (2)} \quad 3 \times 3 + b = 2 \rightarrow b = -7$$

5. $x^3 - 2x^2 + a = (x+2)Q(x) + 3$

$$(-2)^3 - 2(-2)^2 + a = (-2+2)Q(x) + 3$$

$$-8 - 8 + a = 3$$

$$\therefore a = 19$$

6. From the remainder theorem,

$$P(a) = aa^3 + ba^2 + ca + d = p \dots (1)$$

$$P(-a) = -aa^3 + ba^2 - ca + d = -p \dots (2)$$

$$(1) + (2) \quad 2ba^2 + 2d = 0$$

$$\therefore a^2 = -\frac{d}{b}$$

$$(1) - (2) \quad 2a\alpha^3 + 2c\alpha = 2p$$

$$a\alpha^3 + c\alpha = p$$

$$\alpha(a\alpha^2 + c) = p$$

$$\alpha\left(a \times -\frac{d}{b} + c\right) = p$$

$$\alpha\left(\frac{bc - ad}{b}\right) = p$$

$$\therefore \alpha = \frac{pb}{bc - ad}$$

7. When $x = 1$, $P(1) = (1-p)^3 + q = 0 \dots (1)$

If the remainder is -7 when divided by x then

$$P(0) = -7$$

$$\therefore (-p)^3 + q = -7 \dots (2)$$

$$q = p^3 - 7$$

$$\text{In (1)} \quad 1 - 3p + 3p^2 - p^3 + q = 0$$

$$1 - 3p + 3p^2 - p^3 + p^3 - 7 = 0$$

$$3p^2 - 3p - 6 = 0$$

$$p^2 - p - 2 = 0$$

$$(p-2)(p+1) = 0$$

$$\therefore p = 2 \text{ or } p = -1$$

$$\text{when } p = 2, \quad q = 1$$

$$\text{when } p = -1, \quad q = -8$$

8. $P(n) = n(n+m) - n(n+m)$

$$= 0$$

$$\therefore (x-n) \text{ is a factor of } P(x)$$

$$P(x) = x(x+m) - n(n+m)$$

$$= x^2 + mx - n^2 - mn$$

$$x + (n+m)$$

$$x - n \sqrt{x^2 + mx - n^2 - mn}$$

$$\frac{x^2 - nx}{x + (n+m)}$$

$$(n+m)x - n^2 - mn$$

$$\frac{(n+m)x - n^2 - mn}{x + (n+m)}$$

$$0$$

Hence the other factor is $x + (n+m)$.

OR factorise

$$x^2 + mx - n^2 - mn = x^2 - n^2 + mx - mn$$

$$= (x-n)(x+n) + m(x-n)$$

$$= (x-n)(x+n+m)$$

$$\begin{aligned}
 9. \quad P(2) &= 2(2)^3 - 3m(2)^2 - 7(2) - 2 \\
 &= 16 - 12m - 16 \\
 &= -12m \\
 P(2) &= 12 \\
 \therefore -12m &= 12 \\
 m &= -1
 \end{aligned}$$

$$\begin{aligned}
 10. \quad & \begin{array}{r} -x^2 + 5x - 11 \\ 2x + 3 \overline{) -2x^3 + 7x^2 - 7x + 1} \\ \underline{-2x^3 - 3x^2} \\ 10x^2 - 7x + 1 \\ \underline{10x^2 + 15x} \\ -22x + 1 \\ \underline{-22x - 33} \\ 34 \end{array} \\
 \therefore \text{The quotient is } & -x^2 + 5x - 11 \\
 \text{The remainder is } & 34
 \end{aligned}$$

$$\begin{aligned}
 11. \quad & \text{For } x-1, \\
 P(1) &= 2+a+b+6=0 \rightarrow a+b=-8 \dots (1) \\
 & \text{For } x+2, \\
 P(-2) &= -16+4a-2b+6=-12 \rightarrow 4a-2b=-2 \\
 & \qquad \qquad \qquad 2a-b=-1 \dots (2) \\
 (1)+(2) \quad & 3a=-9 \rightarrow a=-3 \\
 & \qquad \qquad \qquad b=-5 \\
 \therefore a &= -3, b = -5
 \end{aligned}$$

$$\begin{aligned}
 12. \quad & \begin{array}{r} x^2 + 1 \\ x-1 \overline{) x^3 - x^2 + x - 1} \\ \underline{-x^3 + x^2} \\ x - 1 \\ \underline{-x + 1} \\ 0 \end{array} \\
 \therefore \text{Quotient} &= x^2 + 1 \\
 \text{Remainder} &= 0
 \end{aligned}$$

$$\begin{aligned}
 13. \quad & \begin{array}{r} x^3 + x^2 + 3x - 12 \\ x+2 \overline{) x^4 + 3x^3 + 5x^2 - 6x - 8} \\ \underline{-x^4 - 2x^3} \\ x^3 + 5x^2 \\ \underline{-x^3 - 2x^2} \\ 3x^2 - 6x \\ \underline{-3x^2 - 6x} \\ -12x - 8 \\ \underline{12x + 24} \\ 16 \end{array} \\
 \therefore x^4 + 3x^3 + 5x^2 - 6x - 8 &= (x+2)(x^3 + x^2 + 3x - 12) + 16 \\
 \therefore \text{Quotient} &= x^3 + x^2 + 3x - 12 \\
 \text{Remainder} &= 16
 \end{aligned}$$

$$\begin{aligned}
 14. \quad & \begin{array}{r} 2x^2 - 4x - 2 \\ x^2 - 3x + 1 \overline{) 2x^4 - 10x^3 + 12x^2 + 2x - 3} \\ \underline{-2x^4 + 6x^3 - 2x^2} \\ -4x^3 + 10x^2 + 2x \\ \underline{4x^3 - 12x^2 + 4x} \\ -2x^2 + 6x - 3 \\ \underline{2x^2 - 6x + 2} \\ -1 \end{array} \\
 \therefore \text{Quotient} &= 2x^2 - 4x - 2 \\
 \text{Remainder} &= -1
 \end{aligned}$$

$$\begin{aligned}
 15. \quad & \begin{array}{r} x^2 + 2x - 3 \\ x-2 \overline{) x^3 - 7x - 3} \\ \underline{-x^3 + 2x^2} \\ 2x^2 - 7x \\ \underline{-2x^2 + 4x} \\ -3x - 3 \\ \underline{3x - 6} \\ -9 \end{array} \\
 \therefore \text{Quotient} &= x^2 + 2x - 3 \\
 \text{Remainder} &= -9
 \end{aligned}$$

$$\begin{aligned}
 16. \quad & \begin{array}{r} x^2 - 3x - 2 \\ x^2 + 2 \overline{) x^4 - 3x^3 - 4} \\ \underline{-x^4} -2x^2 \\ -3x^3 - 2x^2 \\ \underline{3x^3} +6x \\ -2x^2 + 6x - 4 \\ \underline{2x^2} +4 \\ 6x \end{array} \\
 x^4 - 3x^3 - 4 &= (x^2 + 2)(x^2 - 3x - 2) + 6x \\
 \therefore \text{Quotient} &= x^2 - 3x - 2 \\
 \text{Remainder} &= 6x
 \end{aligned}$$

$$\begin{aligned}
 17. \quad & x^2 + 1 = 0 \\
 & x = \pm\sqrt{-1} \\
 \therefore \text{Division is required} \\
 & \begin{array}{r} x^2 - 2 \\ x^2 + 1 \overline{) x^4 - x^2 + 1} \\ \underline{-x^4 - x^2} \\ -2x^2 + 1 \\ \underline{2x^2 + 2} \\ 3 \end{array} \\
 \therefore \text{The remainder is } & 3
 \end{aligned}$$

$$\begin{aligned}
 18. \quad & P(x) = x^3 + 3x^2 + 7x - b \\
 P(2) &= 8 + 12 + 14 - b \\
 &= 34 - b \\
 x^3 + 3x^2 + 7x - b &= (x-2)Q(x) + 20 \\
 \therefore 34 - b &= 20 \\
 b &= 14
 \end{aligned}$$

$$19. P(x) = x^3 + 3x^2 + 2x - 6$$

$$P(2) = 8 + 12 + 4 - 6$$

$$= 18$$

$$\text{Now } x^3 + 3x^2 + 2x - 6 = (x - 2)Q(x) + 18$$

$$\therefore \text{ When } x = 2, \text{ RHS} = 18$$

$$\therefore \text{ Remainder} = 18$$

$$20. P(x) = 2x^3 + 7x^2 + ax + b$$

$$P(3) = 54 + 63 + 3a + b$$

$$= 117 + 3a + b$$

$$P(3) = 120$$

$$\therefore 117 + 3a + b = 120$$

$$3a + b = 3 \dots (1)$$

$$P(-1) = -2 + 7 - a + b$$

$$= 5 - a + b$$

$$P(-1) = -8$$

$$\therefore 5 - a + b = -8$$

$$-a + b = -13 \dots (2)$$

$$(1) - (2)$$

$$4a = 16$$

$$\therefore a = 4 \text{ and } b = -9$$

$$21. P(-1) = 1 - 5 + a + b = 7$$

$$a + b = 11 \dots (1)$$

$$Q(-1) = a - b - 1 = -6$$

$$a - b = -5 \dots (2)$$

$$(1) + (2) \quad 2a = 6$$

$$\therefore a = 3 \text{ and } b = 8$$

$$22. P(x) = ax^2 + bx + c$$

$$P(p) = ap^2 + bp + c = q \dots (1)$$

$$P(-p) = ap^2 - bp + c = r \dots (2)$$

$$(1) - (2) \quad 2bp = q - r$$

$$\therefore p = \frac{q - r}{2b}$$

$$23. P(2) = 24 - 4a - 2b + 1 = 15$$

$$-4a - 2b = -10$$

$$2a + b = 5 \dots (1)$$

$$P(1) = 3 - a - b + 1 = 0$$

$$a + b = 4 \dots (2)$$

$$(1) - (2) \quad \therefore a = 1 \text{ and } b = 3$$

$$24. P(-2) = -8 + 12 + 2m + n = 9$$

$$2m + n = 5 \dots (1)$$

$$P(3) = 27 + 27 - 3m + n = 49$$

$$-3m + n = -5 \dots (2)$$

$$(1) - (2) \quad 5m = 10$$

$$\therefore m = 2 \text{ and } n = 1$$

$$25. P(-2) = -8 + 8 - 6 + q$$

$$= q - 6$$

$$Q(-2) = -8 + 4 + 9 = 5$$

$$\therefore q - 6 = 5$$

$$q = 11$$

$$26. \text{ For } x = 3$$

$$P(3) = 3^4 - 5 \times 3^3 + 6 \times 3^2 - 3a + b$$

$$= 81 - 135 + 54 - 3a + b$$

$$= -3a + b$$

$$P(3) = 20$$

$$\therefore -3a + b = 20 \dots (1)$$

$$\text{For } x = -2$$

$$P(-2) = (-2)^4 - 5(-2)^3 + 6(-2)^2 + 2a + b$$

$$= 16 + 40 + 24 + 2a + b$$

$$= 2a + b + 80$$

$$P(-2) = 30$$

$$\therefore 2a + b + 80 = 30$$

$$2a + b = -50 \dots (2)$$

$$-3a + b = 20 \dots (1)$$

$$(2) - (1) \quad 5a = -70$$

$$a = -14$$

$$\text{In (2)} \quad -28 + b = -50$$

$$b = -22$$

$$\therefore a = -14 \text{ and } b = -22$$

$$27. P(x) = x^3 - 4x^2 + k$$

$$P(2) = 8 - 16 + k$$

$$= k - 8$$

$$P(2) = 4$$

$$\therefore k - 8 = 4$$

$$k = 12$$

$$28. P(x) = (x^2 - 9)Q(x) + x + 1$$

$$P(x) = (x + 3)(x - 3)Q(x) + x + 1$$

$$P(3) = 3 + 1$$

$$= 4$$

∴ The remainder is 4

$$29. P(x) = x^3 - kx^2 + 2x$$

$$P(2) = 8 - 4k + 4 = 4$$

$$-4k = -8$$

$$k = 2$$

$$P(x) = x^3 - 2x^2 + 2x$$

$$P(1) = 1 - 2 + 2$$

$$= 1$$

∴ The remainder is 1

$$30. P(2) = 8 - 2a + b = 4$$

$$-2a + b = -4 \dots (1)$$

$$P(-2) = -8 + 2a + b = 8$$

$$2a + b = 16 \dots (2)$$

$$(1) + (2) \quad 2b = 12$$

$$b = 6$$

$$\text{In (2)} \quad 2a + 6 = 16$$

$$2a = 10$$

$$a = 5$$

∴ $a = 5$ and $b = 6$

31.

$$\text{i) } P(1) = 0 + 0 + b = 10$$

$$\therefore b = 10$$

$$\text{ii) } P(-2) = -3a + 10 = -2$$

$$-3a = -12$$

$$a = 4$$

$$P(x) = (x - 1)(x + 2)Q(x) + 4(x - 1) + 10$$

∴ The remainder is $4x + 6$

$$32. \begin{array}{r} x^2 - 9x + 2 \\ x^2 + x \overline{) x^4 - 8x^3 - 7x^2 + 3} \\ \underline{-x^4 - x^3} \\ -9x^3 - 7x^2 \\ \underline{9x^3 + 9x^2} \\ 2x^2 \\ \underline{-2x^2 - 2x} \\ -2x + 3 \end{array}$$

$$\therefore -2x + 3 = ax + 3$$

$$\therefore a = -2 \quad \therefore C$$

$$33. \text{ Let } P(x) = x^3 - 6x^2$$

$$P(-3) = (-3)^3 - 6(-3)$$

$$= -9 \quad \therefore A$$

$$34. \text{ Let } P(x) = 2x^3 - 10x^2 + 6x + 2$$

$$P(2) = 2(2)^3 - 10 \times 2^2 + 6 \times 2 + 2$$

$$= 16 - 40 + 12 + 2$$

$$= -10 \quad \therefore B$$

$$35. \begin{array}{r} x^2 + 4x + 5 \\ x - 2 \overline{) x^3 + 2x^2 - 3x - 7} \\ \underline{-x^3 + 2x^2} \\ 4x^2 - 3x \\ \underline{-4x^2 + 8x} \\ 5x - 7 \\ \underline{-5x + 10} \\ 3 \end{array}$$

$$36. P(x) = -x^3 - 2x^2 - 3x + 8$$

$$P(-2) = -(-2)^3 - 2(-2)^2 - 3(-2) + 8$$

$$= 8 - 8 + 6 + 8$$

$$= 14$$

$$= D$$

37. The degree of $2x + 5$ is 1.

∴ The degree of the remainder must always be less than the degree of the divisor. Hence Not A on B.

$$\text{If } P(x) = Q(x)A(x) + R(x)$$

$$\text{Let } P(x) = x^5 + x^3 - x^2 + 2x + 4$$

$$Q(x) = x^3 - 1, A(x) = x^2 + 1, R(x) = 2x + 5$$

$$\therefore \text{ i.e. } x^5 + x^3 - x^2 + 2x + 4 = (x^3 - 1)(x^2 + 1) + 2x + 5$$

The degree of $Q(x)$ is 3 and the degree of $R(x)$ is 1.

OR if $A(x)$ is the divisor then the degree of $A(x)$ is 2.

∴ The degree of the divisor could be 2 or 3. ∴ D

$$38. P(-1) = (-1)^3 + a(-1)^2 + b(-1) - 12$$

$$= -1 + a - b - 12$$

$$= a - b - 13$$

$$\therefore a - b - 13 = 0$$

$$a - b = 13 \dots (1)$$

$$P(2) = (2)^3 + a(2)^2 + 2b - 12$$

$$= 8 + 4a + 2b - 12$$

$$= 4a + 2b - 4$$

$$\therefore 4a + 2b - 4 = -18$$

$$4a + 2b = -14$$

$$2a + b = -7 \dots (2)$$

$$(1) + (2) \quad 3a = 6$$

$$a = 2$$

$$\text{In (1)} \quad 2 - b = 13$$

$$b = -11$$

$$\therefore a = 2, b = -11$$

Factor Theorem

39. $P(x) = x^3 - 2kx - 3$

By the factor theorem $P(-3) = 0$

$$P(-3) = -27 + 6k - 3 = 0$$

$$6k = 30$$

$$\therefore k = 5$$

40. $P(x) = x^3 + ax + b$

$$P(3) = 3^3 + 3a + b = 0 \rightarrow 3a + b = -27 \dots (1)$$

$$P(1) = 1^3 + a + b = 0 \rightarrow a + b = -1 \dots (2)$$

$$(1) - (2) \quad 2a = -26$$

$$a = -13$$

$$\text{In (2)} \quad -13 + b = -1$$

$$b = 12$$

41. $P(x) = x^3 - 13x + a$

$$P(3) = 3^3 - 13 \times 3 + a = 0$$

$$27 - 39 + a = 0$$

$$a = 12$$

42. $P(x) = x^3 - bx + 3$

$$P(-1) = (-1)^3 - b \times (-1) + 3 = 0$$

$$-1 + b + 3 = 0$$

$$b = -2$$

43. $P(x) = 2x^3 + kx^2 + 5x - 2$

$$P(-2) = 2(-2)^3 + k(-2)^2 + 5 \times (-2) - 2 = 0$$

$$-16 + 4k - 10 - 2 = 0$$

$$4k - 28 = 0$$

$$k = 7$$

44. $P(x) = x^3 + ax^2 + bx - 4$

$$P(1) = 1 + a + b - 4 = 0 \rightarrow a + b = 3 \dots (1)$$

$$P(-1) = -1 + a - b - 4 = 0 \rightarrow a - b = 5 \dots (2)$$

$$(1) + (2) \quad 2a = 8$$

$$a = 4$$

$$\text{In (1)} \quad 4 + b = 3$$

$$b = -1$$

45. $P(-1) = -3 + 10 - 9 + 2 = 0$

$\therefore x + 1$ is a factor.

$$\begin{array}{r} 3x^2 + 7x + 2 \\ x+1 \overline{) 3x^3 + 10x^2 + 9x + 2} \\ \underline{-3x^3 \quad -3x^2} \\ 7x^2 + 9x \\ \underline{-7x^2 - 7x} \\ 2x + 2 \\ \underline{-2x - 2} \\ 0 \end{array}$$

$$P(x) = (x+1)(3x^2 + 7x + 2)$$

$$= (x+1)(3x+1)(x+2)$$

46. $P(1) = 1 + a + b - 4 = 0$

$$a + b = 3 \dots (1)$$

$$P(-1) = -1 + a - b - 4 = 0$$

$$a - b = 5 \dots (2)$$

$$(1) + (2) \quad 2a = 8 \rightarrow a = 4 \text{ and } b = -1$$

$$\therefore x^3 + 4x^2 - x - 4 = (x-1)(x+1)(x+c)$$

$$\text{Let } x = 0 \quad -4 = -c$$

$$\therefore c = 4$$

$\rightarrow$ the other factor is $x + 4$

47. Let $A(x) = x^2 - 3x + 2$

$$= (x-2)(x-1)$$

$\therefore (x-2)$ and $(x-1)$ are factors of $P(x)$

$$P(1) = 1 + 1 + 1 + a + b = 0$$

$$a + b = -3 \dots (1)$$

$$P(2) = 16 + 8 + 4 + 2a + b = 0$$

$$2a + b = -28 \dots (2)$$

$$(1) - (2) \quad -a = 25 \rightarrow a = -25$$

$$\text{put into (2)} \quad b = 22$$

48. $p(x) = x^3 + 5x^2 + 8x + 4$

Try $x = -1$

$$p(-1) = -1 + 5 - 8 + 4 = 0$$

$\therefore x + 1$ is a factor

$$\therefore x^3 + 5x^2 + 8x + 4 = (x+1)(x^2 + kx + 4)$$

Let $x = 1$

$$1 + 5 + 8 + 4 = 2(1 + k + 4)$$

$$18 = 2(5 + k)$$

$$9 = 5 + k \rightarrow k = 4$$

$$(x+1)(x^2 + 4x + 4) = 0$$

$$(x+1)(x+2)^2 = 0$$

$$\rightarrow x = -1, x = -2, x = -2$$

49. $P(-2) = -8 + 12 + 20 - 24 = 0$

$\therefore x + 2$ is a factor

$$\begin{array}{r} x^2 + x - 12 \\ x+2 \overline{) x^3 + 3x^2 - 10x - 24} \\ \underline{-x^3 - 2x^2} \\ x^2 - 10x \\ \underline{-x^2 - 2x} \\ -12x - 24 \\ \underline{12x + 24} \\ 0 \end{array}$$

$$\therefore P(x) = (x+2)(x^2 + x - 12)$$

$$= (x+2)(x+4)(x-3)$$

50. $P(-2) = -8 - 12 + 16 + 4 = 0$

$\therefore x + 2$ is a factor

$$\begin{array}{r} x^2 - 5x + 2 \\ x+2 \overline{) x^3 - 3x^2 - 8x + 4} \\ \underline{-x^3 - 2x^2} \\ -5x^2 - 8x \\ \underline{5x^2 + 10x} \\ 2x + 4 \\ \underline{-2x - 4} \\ 0 \end{array}$$

$\therefore P(x) = (x + 2)(x^2 - 5x + 2)$

Consider $x^2 - 5x + 2 = 0$

$$x = \frac{5 \pm \sqrt{25 - 4 \times 2 \times 1}}{2}$$

$$x = \frac{5 \pm \sqrt{17}}{2}$$

$\therefore x = -2, \frac{5 + \sqrt{17}}{2}, \frac{5 - \sqrt{17}}{2}$

51.

$$\begin{array}{r} x^3 + x^2 + 3x - 12 \\ x+2 \overline{) x^4 + 3x^3 + 5x^2 - 6x - 8} \\ \underline{-x^4 - 2x^3} \\ x^3 + 5x^2 \\ \underline{-x^3 - 2x^2} \\ 3x^2 - 6x \\ \underline{-3x^2 - 6x} \\ -12x - 8 \\ \underline{12x + 24} \\ 16 \end{array}$$

$\therefore x^4 + 3x^3 + 5x^2 - 6x - 8 = (x + 2)(x^3 + x^2 + 3x - 12) + 16$

$\therefore$ Quotient $= x^3 + x^2 + 3x - 12$

Remainder $= 16$

52.

$$\begin{array}{r} 2x^2 - 4x - 2 \\ x^2 - 3x + 1 \overline{) 2x^4 - 10x^3 + 12x^2 + 2x - 3} \\ \underline{-2x^4 + 6x^3 - 2x^2} \\ -4x^3 + 10x^2 + 2x \\ \underline{4x^3 - 12x^2 + 4x} \\ -2x^2 + 6x - 3 \\ \underline{2x^2 - 6x + 2} \\ -1 \end{array}$$

$\therefore$ Quotient $= 2x^2 - 4x - 2$

Remainder $= -1$

53.

$$\begin{array}{r} x^2 + 2x - 3 \\ x-2 \overline{) x^3 - 7x - 3} \\ \underline{-x^3 + 2x^2} \\ 2x^2 - 7x \\ \underline{-2x^2 + 4x} \\ -3x - 3 \\ \underline{3x - 6} \\ -9 \end{array}$$

$\therefore$ Quotient $= x^2 + 2x - 3$

Remainder $= -9$

54.

$$\begin{array}{r} x^2 - 3x - 2 \\ x^2 + 2 \overline{) x^4 - 3x^3 - 4} \\ \underline{-x^4} - 4 \\ -3x^3 - 2x^2 \\ \underline{3x^3} + 6x \\ -2x^2 + 6x - 4 \\ \underline{2x^2} + 4 \\ 6x \end{array}$$

$x^4 - 3x^3 - 4 = (x^2 + 2)(x^2 - 3x - 2) + 6x$

$\therefore$ Quotient $= x^2 - 3x - 2$

Remainder $= 6x$

55. $x^2 + 1 = 0$

$x = \pm \sqrt{-1}$

$\therefore$ Division is required

$$\begin{array}{r} x^2 - 2 \\ x^2 + 1 \overline{) x^4 - x^2 + 1} \\ \underline{-x^4 - x^2} \\ -2x^2 + 1 \\ \underline{2x^2 + 2} \\ 3 \end{array}$$

$\therefore$ The remainder is 3

56.

$$\begin{array}{r} -x^2 + 5x - 11 \\ 2x + 3 \overline{) 2x^3 + 7x^2 - 7x + 1} \\ \underline{2x^3 + 3x^2} \\ 10x^2 - 7x \\ \underline{-10x^2 - 15x} \\ -22x + 1 \\ \underline{22x + 33} \\ 34 \end{array}$$

$\therefore$ The quotient is $-x^2 + 5x - 11$

The remainder is 34

57. $P(-1) = -1 - 1 + 10 - 8 = 0$

$\therefore x + 1$ is a factor.

$P(x) = (x + 1)(x^2 + bx + c)$

$\therefore (x + 1)(x^2 + bx + c) \equiv x^3 - x^2 - 10x - 8$

By inspection $c = -8$

Let $x = 1 \rightarrow 2(1 + b - 8) = 1 - 1 - 10 - 8$

$2b - 14 = -18$

$\therefore b = -2$

$\therefore P(x) = (x + 1)(x^2 - 2x - 8)$

$= (x + 1)(x - 4)(x + 2)$

58. $P(3) = 27 - 54 + 3p + q = 0$

$3p + q = 27 \dots (1)$

$P(-2) = -8 - 24 - 2p + q = 0$

$-2p + q = 32 \dots (2)$

(1) - (2) $5p = -5$

$\therefore p = -1$ and $q = 30$

59. $P(x) = x^3 + 2^3 = (x+2)(x^2 - 2x + 4)$ (sum of two cubes)

Consider $x^2 - 2x + 4 = 0$

$$\Delta = b^2 - 4ac$$

$$= 4 - 4 \times 1 \times 4$$

$$= -12$$

$\therefore$ no real roots

$\therefore$ no linear factors of $x^2 - 2x + 4$ over the real number field.

$\therefore x+2$ is the only linear factor over the real number field.

60. $P(2) = 24 - 4a - 2b + 1 = 15$

$$-4a - 2b = -10$$

$$2a + b = 5 \dots\dots(1)$$

$$P(1) = 3 - a - b + 1 = 0$$

$$a + b = 4 \dots\dots(2)$$

$$(1) - (2)$$

$$\therefore a = 1 \text{ and } b = 3$$

61. $P(-2) = -8 + 12 + 2m + n = 9$

$$2m + n = 5 \dots\dots(1)$$

$$P(3) = 27 + 27 - 3m + n = 49$$

$$-3m + n = -5 \dots\dots(2)$$

$$(1) - (2) \quad 5m = 10$$

$$\therefore m = 2 \text{ and } n = 1$$

62. $P(-2) = -8 + 8 - 6 + q$

$$= q - 6$$

$$Q(-2) = -8 + 4 + 9$$

$$= 5$$

$$\therefore q - 6 = 5$$

$$q = 11$$

63. For $x = 3$

$$P(3) = 3^4 - 5 \times 3^3 + 6 \times 3^2 - 3a + b$$

$$= 81 - 135 + 54 - 3a + b$$

$$= -3a + b$$

$$P(3) = 20$$

$$\therefore -3a + b = 20 \dots\dots(1)$$

For $x = -2$

$$P(-2) = (-2)^4 - 5(-2)^3 + 6(-2)^2 + 2a + b$$

$$= 16 + 40 + 24 + 2a + b$$

$$= 2a + b + 80$$

$$P(-2) = 30$$

$$\therefore 2a + b + 80 = 30$$

$$2a + b = -50 \dots(2)$$

$$-3a + b = 20 \dots\dots(1)$$

$$(2) - (1) \quad 5a = -70$$

$$a = -14$$

In (2) $-28 + b = -50$

$$b = -22$$

$$\therefore a = -14 \text{ and } b = -22$$

64. $P(x) = x^3 - 4x^2 + k$

$$P(2) = 8 - 16 + k$$

$$= k - 8$$

$$P(2) = 4$$

$$\therefore k - 8 = 4$$

$$k = 12$$

65. $P(x) = (x^2 - 9)Q(x) + x + 1$

$$P(x) = (x+3)(x-3)Q(x) + x + 1$$

$$P(3) = 3 + 1$$

$$= 4$$

$\therefore$ The remainder is 4

66. $P(2) = 8 - 2a + b = 4$

$$-2a + b = -4 \dots\dots(1)$$

$$P(-2) = -8 + 2a + b = 8$$

$$2a + b = 16 \dots(2)$$

$$(1) + (2) \quad 2b = 12$$

$$b = 6$$

In (2) $2a + 6 = 16$

$$2a = 10$$

$$a = 5$$

$$\therefore a = 5 \text{ and } b = 6$$

67.

i) $P(1) = 0 + 0 + b = 10$

$$\therefore b = 10$$

ii) $P(-2) = -3a + 10 = -2$

$$-3a = -12$$

$$a = 4$$

$$P(x) = (x-1)(x+2)Q(x) + 4(x-1) + 10$$

$\therefore$ The remainder is $4x + 6$

68.

(i) If $x - 3$ is a factor then $P(3) = 0$

$$P(3) = 27 - 9k + 15 + 12 = 0$$

$$9k = 54$$

$$k = 6$$

(ii) $x^3 - 6x + 5x + 12 \equiv (x-3)(x^2 + ax + 6)$

$$\text{Let } x = 0 \quad 12 \equiv -3 \times b \rightarrow b = -4$$

$$\text{Let } x = 1 \quad 1 - 6 + 5 + 12 = (1 - 3)(1 + a - 4) \\ 12 = -2(a - 3) \rightarrow a = -3$$

$$P(x) = (x - 3)(x^2 - 3x - 4) \\ = (x - 3)(x - 4)(x + 1) \\ \therefore \text{zeros are } 3, 4, -1.$$

69. For $x - 2$:

$$P(2) = 2^3 - 5(2)^2 + 11 \times 2 - 10 \\ = 8 - 20 + 22 - 10 \\ = 0$$

$\therefore x - 2$ is a factor. $\therefore A$

70. $P(1) = 1 - 2 - 5 + 6$

$$= 0$$

$\therefore x = 1$ is a zero.

$$\begin{array}{r} x^2 - x - 6 \\ x-1 \overline{) x^3 - 2x^2 - 5x + 6} \\ \underline{-x^3 + x^2} \\ -x^2 - 5x \\ \underline{x^2 - x} \\ -6x + 6 \\ \underline{6x - 6} \\ 0 \end{array}$$

$$P(x) = (x - 1)(x^2 - x - 6) \\ = (x - 1)(x - 3)(x + 2)$$

Other zeros are 3, -2.

71. Let the other zero = β

$$\alpha\beta \times -1 = \frac{-q}{q}$$

$$\alpha\beta = 1$$

$$\rightarrow \beta = \frac{1}{\alpha} \quad \therefore C$$

72.

$$(i) P(2) = (2)^3 + 3(2)^2 - 13(2) + 6 \\ = 8 + 12 - 26 + 6 \\ = 0$$

$$(ii) x^3 + 3x^2 - 13x + 6 = (x - 2)(x^2 + bx + c)$$

$$\text{Let } x = 0$$

$$6 = (-2)(c)$$

$$c = -3$$

$$\text{Let } x = 1$$

$$1 + 3 - 13 + 6 = (-1)(1 + b - 3)$$

$$-3 = -b + 2$$

$$b = 5$$

$$\therefore P(x) = (x - 2)(x^2 + 5x - 3)$$

Roots and Coefficients

$$73. 1 + \alpha + \frac{1}{\alpha} = -a \quad \dots (1)$$

$$1 \times \alpha + 1 \times \frac{1}{\alpha} + \alpha \times \frac{1}{\alpha} = b$$

$$\alpha + \frac{1}{\alpha} + 1 = b \quad \dots (2)$$

$$(2) - (1) \quad b + a = 0$$

74. Let the roots be $\alpha, -\alpha$ and β

$$\therefore \alpha + (-\alpha) + \beta = -\frac{-2}{2} = 1 \rightarrow \beta = 1$$

$$P(x) = 2x^3 - 2x^2 - kx + 6$$

$$P(1) = 2 - 2 - k + 6 = 0$$

$$\therefore k = 6$$

75. Let the roots be α, γ and β

$$\therefore \alpha = \beta + \gamma$$

$$\alpha + \beta + \gamma = 6 \rightarrow 2\alpha = 6 \rightarrow \alpha = 3$$

$$x^2 - 3x + 2$$

$$x - 3 \sqrt{x^3 - 6x^2 + 11x - 6}$$

$$\underline{x^3 - 3x^2}$$

$$-3x^2 + 11x - 6$$

$$\underline{-3x^2 + 9x}$$

$$2x - 6$$

$$\underline{2x - 6}$$

$$0$$

$$P(x) = (x - 3)(x^2 - 3x + 2)$$

$$= (x - 3)(x - 2)(x - 1)$$

$\therefore$ roots are 1, 2, 3

76. $\alpha + \beta = -4, \quad \alpha\beta = -2$

$$\text{Quadratic is of the form: } x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)x + \frac{\alpha}{\beta} \times \frac{\beta}{\alpha} = 0$$

$$x^2 - \left(\frac{\alpha^2 + \beta^2}{\alpha\beta}\right)x + 1 = 0$$

$$(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2 \rightarrow \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= 16 - 2 \times -2$$

$$= 20$$

$$\therefore x^2 - \frac{20}{-2}x + 1 = 0$$

$$x^2 + 10x + 1 = 0$$

77. Let the roots be $a - d, a, a + d$

$$a - d + a + a - d = 6 \rightarrow a = 2$$

$$P(x) = x^3 - 6x^2 + 3x + m$$

$$P(2) = 8 - 24 + 6 + m = 0$$

$$\therefore m = 10$$

78. Let the roots be $\alpha, \frac{1}{\alpha}, \gamma$, since $\alpha \times \frac{1}{\alpha} = 1$

$$\therefore \alpha \times \frac{1}{\alpha} \times \gamma = -\frac{4}{2} \rightarrow \gamma = -2$$

$$\begin{aligned}\therefore 2(-2)^3 - k(-2)^2 - 8\gamma - 2 + 4 &= 0 \\ -16 - 4k + 16 + 4 &= 0 \\ -4k + 4 &= 0 \\ k &= 1\end{aligned}$$

79. $x^3 - 7x^2 + 16x + k = 0$

Let the roots be α, α, β

$$\therefore \alpha + \alpha + \beta = 7$$

$$2\alpha + \beta = 7 \dots (1)$$

$$\alpha^2 + 2\beta + \alpha\beta = 16$$

$$\alpha^2 + 2\alpha\beta = 16 \dots (2)$$

Sub (1) into (2)

$$\alpha^2 + 2\alpha(7 - 2\alpha) = 16$$

$$\alpha^2 + 14\alpha - 4\alpha^2 = 16$$

$$3\alpha^2 - 14\alpha + 16 = 0$$

$$(3\alpha - 8)(\alpha - 2) = 0$$

Integral root: $\alpha = 2, \beta = 3$

Roots are 2, 2, 3

$$\therefore k = -12$$

80. $\alpha + \beta = -8$ and $\alpha\beta = -5$

$$\begin{aligned}\frac{\alpha}{\beta} + \frac{\beta}{\alpha} &= \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} \\ &= \frac{64 + 10}{-5} \\ &= -\frac{74}{5}\end{aligned}$$

$$\frac{\alpha}{\beta} \times \frac{\beta}{\alpha} = 1$$

$\therefore$ Quadratic equation is:

$$x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)x + \alpha\beta = 0$$

$$x^2 + \frac{74x}{5} + 1 = 0$$

$$5x^2 + 74x + 5 = 0$$

81. Let the roots be α, β, γ

$$\therefore \alpha = \beta + \gamma$$

$$\begin{aligned}\therefore \alpha + \beta + \gamma &= \alpha + \alpha \\ &= 2\alpha\end{aligned}$$

$$\therefore 2\alpha = -p \rightarrow \alpha = -\frac{p}{2}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 0$$

$$\alpha(\beta + \gamma) + \beta\gamma = 0$$

$$\alpha^2 + \beta\gamma = 0 \rightarrow \beta\gamma = -\frac{p^2}{4}$$

$$\alpha\beta\gamma = -1 \rightarrow \therefore -\frac{p}{2} \times -\frac{p^2}{4} = -1$$

$$\frac{p^3}{8} = -1$$

$$p^3 = -8$$

$$\therefore p = -2$$

82. $-2 + 4 + \alpha = \frac{8}{p}$

$$\alpha = \frac{8}{p} - 2 \dots (1)$$

$$-2 \times 4 \times \alpha = \frac{32}{p}$$

$$\alpha = -\frac{4}{p} \dots (2)$$

$$\therefore \frac{8}{p} - 2 = -\frac{4}{p}$$

$$\frac{8}{p} + \frac{4}{p} = 2$$

$$12 = 2p \rightarrow p = 6$$

$$\text{In (2)} \quad \alpha = -\frac{4}{p} = -\frac{4}{6}$$

$$\therefore \alpha = -\frac{2}{3}$$

83. Let the roots be α, β and γ .

$$\therefore \alpha = \beta + \gamma$$

$$\alpha + \beta + \gamma = -\frac{32}{4}$$

$$2\alpha = -8$$

$$\therefore \alpha = -4$$

$\therefore x + 4$ is a factor

$$\begin{array}{r} 36x^2 - 144x + 655 \\ x+4 \overline{) 36x^3 + 60} \\ \underline{-36x^3 - 144x^2} \\ -144x^2 + 79x \\ \underline{144x^2 + 576x} \\ 655x + 60 \\ \underline{-655x - 2620} \\ -2560 \end{array}$$

$$(x + 4)(4x^2 + 16x + 15) = 0$$

$$(x + 4)(2x + 3)(2x + 5) = 0$$

$$\therefore x = -4, -1.5, -2.5$$

84. Let the roots be $\alpha, \frac{1}{\alpha}, \beta$.

$$\therefore \alpha \times \frac{1}{\alpha} \times \beta = -1$$

$$\therefore \beta = -1$$

$\therefore x + 1$ is a factor of $P(x)$

$$\begin{array}{r} 2x^2 + 5x + 2 \\ x+1 \overline{) 2x^3 + 7x^2 + 7x + 2} \\ \underline{-2x^3 - 2x^2} \\ 5x^2 + 7x \\ \underline{-5x^2 - 5x} \\ 2x + 2 \\ \underline{-2x - 2} \\ 0 \end{array}$$

$$2x^2 + 5x + 2 = (2x + 1)(x + 2)$$

$$\therefore P(x) = (x + 1)(2x + 1)(x + 2)$$

$$\therefore \text{Roots are } -1, -\frac{1}{2}, -2$$

$$85. \quad a + ar + ar^2 = \frac{21}{2}$$

$$a(1 + r + r^2) = \frac{21}{2} \dots (1)$$

$$a \times ar \times ar^2 = \frac{16}{2}$$

$$a^3 r^3 = 8$$

$$ar = 2 \rightarrow a = \frac{2}{r} \dots (2)$$

Substitute (2) into (1)

$$\frac{2}{r}(1 + r + r^2) = \frac{21}{2}$$

$$4(1 + r + r^2) = 21r$$

$$4 + 4r + 4r^2 = 21r$$

$$4r^2 - 17r + 4 = 0$$

$$(4r - 1)(r - 4) = 0$$

$$r = \frac{1}{4} \text{ or } r = 4$$

$$a = \frac{2}{4} = \frac{1}{2}$$

Roots are a, ar and ar^2

$$\therefore \text{Roots are } \frac{1}{2}, 2, 8$$

$$86. \quad \alpha + \beta + \gamma = -\frac{8}{3}, \alpha\beta + \alpha\gamma + \beta\gamma = 0, \alpha\beta\gamma = \frac{1}{3}$$

$$\text{Let } E = \frac{(\beta\gamma + 1)(\gamma\alpha + 1)(\alpha\beta + 1)}{\alpha\beta\gamma}$$

$$= (\alpha\beta\gamma^2 + \beta\gamma + \gamma\alpha + 1)(\alpha\beta + 1) \div \frac{1}{3}$$

$$= 3((\alpha\beta\gamma)^2 + \alpha\beta^2\gamma + \alpha^2\beta\gamma + \alpha\beta + \alpha\beta\gamma^2 + \beta\gamma + \gamma\alpha + 1)$$

$$= 3((\alpha\beta\gamma)^2 + \alpha\beta\gamma(\alpha + \beta + \gamma) + \alpha\beta + \beta\gamma + \gamma\alpha + 1)$$

$$= 3\left(\frac{1}{9} + \frac{1}{3} \times -\frac{8}{3} + 0 + 1\right)$$

$$= \frac{2}{3}$$

$$87. \quad \alpha + \beta + \gamma + \delta = 2$$

$$\text{Let } \gamma + \delta = 0$$

$$\therefore \alpha + \beta = 2 \dots (1)$$

$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = -3$$

$$\alpha\beta + \alpha(\gamma + \delta) + \beta(\gamma + \delta) + \gamma\delta = -3$$

$$\therefore \alpha\beta + \gamma\delta = -3 \dots (2)$$

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -4$$

$$\alpha\beta(\gamma + \delta) + \alpha\gamma\delta + \beta\gamma\delta = -4$$

$$\alpha\gamma\delta + \beta\gamma\delta = -4$$

$$\gamma\delta(\alpha + \beta) = -4$$

$$\therefore 2\gamma\delta = -4 \quad (\text{from (1)})$$

$$\gamma\delta = -2 \dots (3)$$

Substituting (3) into (2)

$$\alpha\beta - 2 = -3$$

$$\alpha\beta = -1 \rightarrow \beta = -\frac{1}{\alpha}$$

$$\text{In (1)} \quad \alpha - \frac{1}{\alpha} = 2$$

$$\alpha^2 - 1 = 2\alpha$$

$$\alpha^2 - 2\alpha - 1 = 0$$

$$\alpha = \frac{2 \pm \sqrt{4 - 4 \times 1 \times -1}}{2} = \frac{2 \pm \sqrt{8}}{2}$$

$$= 1 \pm \sqrt{2}$$

$$\therefore \alpha = 1 + \sqrt{2}, \beta = 1 - \sqrt{2}$$

$$\gamma\delta = -2$$

$$\gamma + \delta = 0 \rightarrow \gamma = -\delta$$

$$\therefore -\delta^2 = -2 \rightarrow \delta = \pm\sqrt{2}$$

$$\therefore \text{Roots are: } 1 + \sqrt{2}, 1 - \sqrt{2}, \sqrt{2}, -\sqrt{2}$$

$$88. \quad \text{Let } \alpha = a - d, \beta = a \text{ and } \gamma = a + d$$

$$\therefore \alpha + \beta + \gamma = 3a$$

$$\therefore 3a = \frac{36}{8} \rightarrow a = \frac{3}{2}$$

$$\alpha\beta\gamma = (a - d)a(a + d) = -\frac{21}{8}$$

$$\left(\frac{3}{2} - d\right)\frac{3}{2}\left(\frac{3}{2} + d\right) = -\frac{21}{8}$$

$$\frac{9}{4} - d^2 = -\frac{7}{4}$$

$$d^2 = 4 \rightarrow d = \pm 2$$

$$\text{When } d = 2 \text{ and } a = \frac{3}{2}$$

$$\text{Roots are } -\frac{1}{2}, \frac{3}{2}, \frac{7}{2}$$

$$\text{When } d = -2 \text{ and } a = \frac{3}{2}$$

$$\text{Roots are } \frac{7}{2}, \frac{3}{2}, -\frac{1}{2}$$

$$89. \quad \text{If } \alpha, \beta, \gamma \text{ are the roots of } P(x) = x^3 - x^2 + 4x - 1, \text{ then}$$

$\alpha + 1, \beta + 1, \gamma + 1$ are the roots of

$$P(x) = (x - 1)^3 - (x - 1)^2 + 4(x - 1) - 1$$

$$\therefore P(x) = x^3 - 3x^2 + 3x - 1 - (x^2 - 2x + 1) + 4x - 4 - 1$$

$$= x^3 - 4x^2 + 9x - 7$$

$$\therefore (\alpha + 1)(\beta + 1)(\gamma + 1) = 7$$

90. Let the roots be α, β, γ

$$\therefore \alpha = \beta + \gamma$$

$$\alpha + \beta + \gamma = 8$$

$$2\alpha = 8 \rightarrow \alpha = 4$$

$$\alpha\beta\gamma = 12$$

$$\therefore 4\beta\gamma = 12$$

$$\beta\gamma = 3 \dots (1)$$

$$\alpha + \beta + \gamma = 8$$

$$4 + \beta + \gamma = 8$$

$$\beta + \gamma = 4 \dots (2)$$

$$\text{From (1), } \gamma = \frac{3}{\beta}$$

Substituting in (2)

$$\beta + \frac{3}{\beta} = 4$$

$$\beta^2 + 3 = 4\beta$$

$$\beta^2 - 4\beta + 3 = 0$$

$$(\beta - 3)(\beta - 1) = 0$$

$$\beta = 3, \gamma = 1 \text{ or } \beta = 1, \gamma = 3$$

$\therefore$ Roots are 1, 3 and 4

91. Let the roots be $\alpha, \frac{1}{\alpha}, \beta$

$$\therefore \alpha \times \frac{1}{\alpha} \times \beta = -\frac{4}{2}$$

$$\therefore \beta = -2$$

$$P(x) = 2x^3 + x^2 - kx + 4$$

$$P(-2) = -16 + 4 + 2k + 4 = 0$$

$$2k = 8$$

$$\therefore k = 4$$

92. Let the roots be $\alpha, \beta, \gamma, \delta$

$$\alpha + \beta + \gamma + \delta = -4$$

$$\text{Now } \gamma + \delta = 0$$

$$\therefore \alpha + \beta = -4$$

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = 16$$

$$\alpha\beta(\gamma + \delta) + \alpha\gamma\delta + \beta\gamma\delta = 16$$

$$\therefore \alpha\gamma\delta + \beta\gamma\delta = 16$$

$$\text{Now } \gamma + \delta = 0 \rightarrow \gamma = -\delta$$

$$\therefore -\alpha\delta^2 - \beta\delta^2 = 16$$

$$-\delta^2(\alpha + \beta) = 16$$

$$-\delta^2 \times -4 = 16$$

$$\therefore \delta = \pm 4$$

$$\therefore \gamma = 2 \text{ and } \delta = -2$$

$$\alpha\beta\gamma\delta = -12$$

$$\alpha\beta \times 2 \times -2 = -12$$

$$\therefore \alpha\beta = 3 \dots (1)$$

$$\alpha + \beta = -4$$

$$\beta = -4 - \alpha \dots (2)$$

$$\text{In (1) } \alpha(-4 - \alpha) = 3$$

$$-4\alpha - \alpha^2 = 3$$

$$\alpha^2 + 4\alpha + 3 = 0$$

$$(\alpha + 1)(\alpha + 3) = 0$$

$$\therefore \alpha = -1 \text{ or } \alpha = -3 \rightarrow \beta = -3 \text{ or } \beta = -1$$

$\therefore$ The roots are 2, -2, -1, -3

93. Let the roots be $\alpha, \beta, \gamma, \delta$

$$\therefore \alpha\beta\gamma\delta = -12$$

$$\text{Let } \gamma\delta = 6$$

$$\therefore \alpha\beta = -2$$

$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -10$$

$$-2\gamma - 2\delta + 6\alpha + 6\beta = -10$$

$$-\gamma - \delta + 3\alpha + 3\beta = -5$$

$$\text{Now } \alpha + \beta + \gamma + \delta = 5 \dots (1)$$

$$3\alpha + 3\beta - \gamma - \delta = -5 \dots (2)$$

$$(1) + (2)$$

$$4\alpha + 4\beta = 0 \rightarrow \alpha = -\beta$$

$$\text{Now } \alpha\beta = -2$$

$$\therefore -\beta^2 = -2 \rightarrow \beta = \pm\sqrt{2}$$

$$\therefore \alpha = \sqrt{2} \text{ and } \beta = -\sqrt{2}$$

$$\text{In (1): } \sqrt{2} - \sqrt{2} + \gamma + \delta = 5$$

$$\gamma + \delta = 5 \dots (3)$$

$$\gamma\delta = 6 \dots (4)$$

Substituting (3) into (4)

$$\gamma(5 - \gamma) = 6$$

$$5\gamma - \gamma^2 = 6$$

$$\gamma^2 - 5\gamma + 6 = 0$$

$$(\gamma - 3)(\gamma - 2) = 0$$

$$\therefore \gamma = 3 \text{ or } \gamma = -2 \rightarrow \delta = -2 \text{ or } \delta = 3$$

$\therefore$ Roots are $\sqrt{2}, -\sqrt{2}, 3, -2$

94. $nx^2 + (2n - m^2)x + n = 0$

$$\therefore x = \frac{(m^2 - 2n) \pm \sqrt{(m^2 - 2n)^2 - 4n^2}}{2n}$$

$$= \frac{(m^2 - 2n) \pm \sqrt{m^4 - 4m^2n + 4n^2 - 4n^2}}{2n}$$

$$= \frac{(m^2 - 2n) \pm \sqrt{m^4 - 4m^2n}}{2n}$$

$$= \frac{(m^2 - 2n) \pm m\sqrt{m^2 - 4n}}{2n}$$

$$\alpha + \beta = -m \text{ and } \alpha\beta = n$$

$$\begin{aligned}\therefore x &= \frac{((\alpha + \beta)^2 - 2\alpha\beta) \mp (\alpha + \beta)\sqrt{(\alpha + \beta)^2 - 4\alpha\beta}}{2n} \\ x &= \frac{(\alpha^2 + \beta^2) \mp (\alpha + \beta)\sqrt{(\alpha - \beta)^2}}{2\alpha\beta} \\ x &= \frac{(\alpha^2 + \beta^2) \mp (\alpha + \beta)(\alpha - \beta)}{2\alpha\beta} \\ x &= \frac{(\alpha^2 + \beta^2) \mp (\alpha^2 - \beta^2)}{2\alpha\beta} \\ x &= \frac{2\alpha^2}{2\alpha\beta} \text{ or } x = \frac{2\beta^2}{2\alpha\beta} \\ x &= \frac{\alpha}{\beta} \text{ or } x = \frac{\beta}{\alpha}\end{aligned}$$

95.

i) $-1 + \alpha - \alpha = -a$

$$\therefore a = 1$$

ii) $P(-1) = -1 + 2 - b + c = 0$

$$-b + c = -1$$

$$\therefore c - b = -1$$

96. $-2 + 4 + \alpha = -\frac{8}{p} \quad -2 \times 4 \times \alpha = \frac{32}{p}$

$$\alpha = -\frac{8}{p} - 2 \dots (1) \quad \alpha = -\frac{4}{p} \dots (2)$$

$$\therefore -\frac{8}{p} - 2 = -\frac{4}{p}$$

$$-\frac{8}{p} + \frac{4}{p} = 2$$

$$-4 = 2p \rightarrow p = -2$$

In (2) $\alpha = -\frac{4}{p}$

$$= -\frac{4}{-2}$$

$$\alpha = 2$$

97. $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma}{\alpha\beta\gamma\delta}$

$$\beta\gamma\delta + \alpha\gamma\delta + \alpha\beta\delta + \alpha\beta\gamma = \frac{6}{2} = 3$$

$$\alpha\beta\gamma\delta = \frac{3}{2}$$

$$\therefore \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = 3 \div \frac{3}{2}$$

$$= 2$$

98. $P(x) = x^3 - 5x^2 + 3x + 9$

Let the roots be α, α and β

$$\therefore \alpha + \alpha + \beta = 5$$

$$2\alpha + \beta = 5$$

$$\beta = 5 - 2\alpha \dots (1)$$

$$\alpha^2 + \alpha\beta + \alpha\beta = 3$$

$$\alpha^2 + 2\alpha\beta = 3 \dots (2)$$

(1) in (2)

$$\alpha^2 + 2\alpha(5 - 2\alpha) = 3$$

$$\alpha^2 + 10\alpha - 4\alpha^2 = 3$$

$$3\alpha^2 - 10\alpha + 3 = 0$$

$$(3\alpha - 1)(\alpha - 3) = 0$$

$$\therefore \alpha = \frac{1}{3} \text{ or } \alpha = 3$$

For $\alpha = \frac{1}{3}, \beta = 5 - \frac{2}{3} = \frac{13}{3}$

$$\therefore \text{Roots are } \frac{1}{3}, \frac{1}{3}, \frac{13}{3}$$

Now $\alpha^2\beta = -9$

$$\frac{1}{3} \times \frac{1}{3} \times \frac{13}{3} \neq -9$$

$$\therefore \frac{1}{3}, \frac{1}{3}, \frac{13}{3} \text{ are not the correct roots}$$

For $\alpha = 3, \beta = 5 - 6 = -1$

$$\therefore \text{Roots are } 3, 3, -1$$

Now $\alpha^2\beta = -9$

$$3 \times 3 \times -1 = -9$$

$$\therefore 3, 3, -1 \text{ are the correct roots.}$$

99.

i) $\alpha + \beta = \frac{4}{2} = 2, \alpha\beta = \frac{3}{2}$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta} = \frac{2}{\frac{3}{2}} = \frac{4}{3}$$

ii) $\frac{\beta}{\alpha} + \frac{\alpha}{\beta} = \frac{\beta^2 + \alpha^2}{\alpha\beta}$

$$\frac{\beta}{\alpha} + \frac{\alpha}{\beta} = \frac{(\beta + \alpha)^2 - 2\beta\alpha}{\alpha\beta}$$

$$= \frac{2^2 - 2 \times \frac{3}{2}}{\frac{3}{2}}$$

$$= \frac{2}{3}$$

100.

i) $\alpha + \beta + \gamma = 4$

ii) $\alpha\beta\gamma = -16$

iii) $\alpha - \alpha + \gamma = 4(\beta - \alpha)$

$$\gamma = 4$$

$$\therefore \text{The third root is } 4$$

$$P(4) = 64 - 64 + 4k + 16 = 0$$

$$4k = -16$$

$$\therefore k = -4$$

iv) From (ii)

$$-\alpha^2 \times 4 = -16$$

$$\alpha^2 = 4 \rightarrow \alpha = \pm 2$$

$$\therefore \text{The other two roots are } -2, 2$$

101. $\alpha + \beta = -p$ and $\alpha\beta = q$

i) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta}$
 $= -\frac{p}{q}$

ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= p^2 - 2q$

iii) $(\alpha + 1)(\beta + 1) = \alpha\beta + \alpha + \beta + 1$
 $= q - p + 1$

102.

i) $\alpha + \beta + \gamma = 0$

ii) $\alpha\beta\gamma = -1$

iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{-2}{-1}$
 $= 2$

103. $\cos 3\theta = \cos(2\theta + \theta) = \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$
 $= (2\cos^2 \theta - 1)\cos \theta - 2\cos \theta \sin^2 \theta$
 $= 2\cos^3 \theta - \cos \theta - 2\cos \theta(1 - \cos^2 \theta)$
 $= 4\cos^3 \theta - 3\cos \theta$

$\therefore 4\cos^3 \theta - 3\cos \theta = -\frac{1}{2}$

$8\cos^3 \theta - 6\cos \theta + 1 = 0$

$\therefore 8x^3 - 6x + 1 = 0$ where $x = \cos \theta$

i) $\cos 3\theta = -\frac{1}{2}$

$\therefore 3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, 2\pi + \frac{2\pi}{3}$

$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$

$\therefore$ the roots are $\cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9}$

now $\cos \frac{8\pi}{9} = -\cos \frac{\pi}{9}$

$\therefore -\cos \frac{\pi}{9} + \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = 0$

$\therefore \cos \frac{\pi}{9} = \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9}$

ii) $-\cos \frac{\pi}{9} \times \cos \frac{2\pi}{9} \times \cos \frac{4\pi}{9} = \frac{-1}{8}$

$\therefore \sec \frac{\pi}{9} \sec \frac{2\pi}{9} \sec \frac{4\pi}{9} = 8$

iii) If $\tan^2 \frac{\pi}{9} + \tan^2 \frac{2\pi}{9} + \tan^2 \frac{4\pi}{9} = 33$

$\Rightarrow \sec^2 \frac{\pi}{9} - 1 + \sec^2 \frac{2\pi}{9} - 1 + \sec^2 \frac{4\pi}{9} - 1 = 33$

$\Rightarrow \sec^2 \frac{\pi}{9} + \sec^2 \frac{2\pi}{9} + \sec^2 \frac{4\pi}{9} = 36$

Let $\alpha = \cos \frac{\pi}{9}, \beta = \cos \frac{2\pi}{9}, \gamma = \cos \frac{4\pi}{9}$

$\therefore \frac{1}{\alpha^2} = \sec^2 \frac{\pi}{9}, \frac{1}{\beta^2} = \sec^2 \frac{2\pi}{9}, \frac{1}{\gamma^2} = \sec^2 \frac{4\pi}{9}$

If α, β, γ are the roots of $8x^3 - 6x + 1 = 0$ then

$\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$ are the roots of:

$8\left(\frac{1}{\sqrt{x}}\right)^3 - 6\left(\frac{1}{\sqrt{x}}\right) + 1 = 0$

$\frac{8}{x\sqrt{x}} - \frac{6}{\sqrt{x}} + 1 = 0$

$8 - 6x + x\sqrt{x} = 0$

$x\sqrt{x} = 6x - 8$

$x^3 = (6x - 8)^2$

$x^3 = 36x^2 - 96x + 64$

$\therefore x^3 - 36x^2 + 96x - 64 = 0$

Now $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{-1 \times -36}{1} = 36$

$\therefore \sec^2 \frac{\pi}{9} + \sec^2 \frac{2\pi}{9} + \sec^2 \frac{4\pi}{9} = 36$

$\therefore \tan^2 \frac{\pi}{9} + \tan^2 \frac{2\pi}{9} + \tan^2 \frac{4\pi}{9} = 33$

104.

$$\begin{aligned} \text{i) } \tan 3\theta &= \frac{\tan 2\theta + \tan \theta}{1 - \tan 2\theta \tan \theta} \\ &= \frac{\frac{2 \tan \theta}{1 - \tan^2 \theta} + \tan \theta}{1 - \frac{2 \tan \theta}{1 - \tan^2 \theta} \times \tan \theta} \\ &= \frac{2 \tan \theta + (\tan \theta)(1 - \tan^2 \theta)}{1 - \tan^2 \theta - 2 \tan^2 \theta} \\ &= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \end{aligned}$$

$$\text{ii) } \tan 3\theta = \frac{3x - x^3}{1 - 3x^2}$$

$$\text{Now } \tan 3\theta = 1,$$

$$\therefore 1 - 3x^2 = 3x - x^3$$

$$x^3 - 3x^2 - 3x + 1 = 0$$

$$\text{Now } \tan 3\theta = 1 \text{ when}$$

$$3\theta = \frac{\pi}{4}, \pi + \frac{\pi}{4}, 2\pi + \frac{\pi}{4}$$

$$\therefore \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}$$

$$\text{Therefore the three roots are } x = \tan \frac{\pi}{12},$$

$$x = \tan \frac{5\pi}{12} \text{ and}$$

$$x = \tan \frac{9\pi}{12} = -1.$$

$$\text{iii) } (\alpha) \quad \tan \frac{\pi}{12} + \tan \frac{5\pi}{12} + -1 = 3$$

$$\therefore \tan \frac{\pi}{12} + \tan \frac{5\pi}{12} = 4$$

$$(\beta) \quad \tan \frac{\pi}{12} \times \tan \frac{5\pi}{12} \times -1 = -1$$

$$\therefore \tan \frac{\pi}{12} \times \tan \frac{5\pi}{12} = 1$$

$$105. P(x) = x^3 + ax^2 - 41x + 42 = 0$$

$$\alpha\beta\gamma = -42 \rightarrow B \text{ on } D$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = -41$$

$$\text{For } 1, -6, 7$$

$$1 \times 7 + 1 \times -6 + -6 \times 7 = -41 \quad \therefore B$$

$$106. \alpha\beta\gamma = \frac{-(-10)}{2} = 5$$

$$\alpha + \beta + \gamma = \frac{-6}{2} = -3$$

$$\therefore \alpha\beta\gamma(\alpha + \beta + \gamma) = 5 \times -3 = -15 \quad \therefore B$$

$$107. \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta} = \frac{\Sigma \alpha\beta\gamma}{\alpha\beta\gamma\delta}$$

$$x^4 - 3x + 6 = 0 \quad a = 1, b = 0, c = 0, d = -3, e = 6$$

$$\Sigma \alpha\beta\gamma = -\frac{d}{a} = 3$$

$$\alpha\beta\gamma\delta = \frac{e}{a} = 6$$

$$\therefore \frac{\Sigma \alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{3}{6} = \frac{1}{2}$$

$$108. \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) = 85$$

$$P(x) = x^3 + bx^2 + cx + d$$

$$\alpha + \beta + \gamma = -b$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = c$$

$$\text{In (1)} \quad b^2 - 2c = 85 \dots (1)$$

$$P'(x) = 3x^2 + 2bx + c$$

$$P'(\alpha) = 3\alpha^2 + 2b\alpha + c$$

$$P'(\beta) = 3\beta^2 + 2b\beta + c$$

$$P'(\gamma) = 3\gamma^2 + 2b\gamma + c$$

$$\therefore P'(\alpha) + P'(\beta) + P'(\gamma) = 3(\alpha^2 + \beta^2 + \gamma^2) + 2b(\alpha + \beta + \gamma) + 3c$$

$$\Rightarrow 3 \times 85 - 2b^2 + 3c = 87$$

$$-2b^2 + 3c = -168$$

$$2b^2 - 3c = 168 \dots (2)$$

$$\text{Sub (1) into (2)}$$

$$2(85 + 2c) - 3c = 168$$

$$170 + 4c - 3c = 168$$

$$\therefore c = -2$$

$$\therefore \alpha\beta + \beta\gamma + \gamma\alpha = -2$$

Multiplicity of Roots

$$109. P(x) = 2x^3 - 15x^2 + 24x + 16$$

$$P'(x) = 6x^2 - 30x + 24 = 0$$

$$6(x^2 - 5x + 4) = 0$$

$$6(x - 4)(x - 1) = 0$$

$$P(1) = 2 - 15 + 24 + 16 \neq 0$$

$$P(4) = 128 - 240 + 96 + 16 = 0$$

$$\therefore x = 4 \text{ is a double root}$$

$$110. P(x) = 4x^3 + 12x^2 - 15x + 4$$

$$P'(x) = 12x^2 + 24x - 15 = 0$$

$$3(4x^2 + 8x - 5) = 0$$

$$3(2x + 5)(2x - 1) = 0$$

$$\therefore x = -2\frac{1}{2} \text{ or } x = \frac{1}{2}$$

$$P\left(\frac{1}{2}\right) = 4 \times \frac{1}{8} + 12 \times \frac{1}{4} - 15 \times \frac{1}{2} + 4 = 0$$

$$\therefore x = \frac{1}{2} \text{ is a double root.}$$

$$111. P(x) = x^4 + 2x^3 - 12x^2 + 14x - 5$$

$$P'(x) = 4x^3 + 6x^2 - 24x + 14$$

$$P''(x) = 12x^2 + 12x - 24 = 0$$

$$12(x^2 + x - 2) = 0$$

$$12(x + 2)(x - 1) = 0$$

$$\therefore x = -2, x = 1$$

$$P(1) = 1 + 2 - 12 + 14 - 5 = 0$$

$$\therefore x = 1 \text{ is a triple root}$$

$$112. P(x) = x^3 - 6x^2 + 9x + k$$

$$P'(x) = 3x^2 - 12x + 9 = 0$$

$$3(x^2 - 4x + 3) = 0$$

$$3(x-3)(x-1) = 0$$

$$\therefore x = 3 \text{ or } x = 1$$

$$\text{At } x = 3 \quad P(3) = 27 - 54 + 27 + k = 0$$

$$\therefore k = 0$$

$$\text{At } x = 1 \quad P(1) = 1 - 6 + 9 + k = 0$$

$$\therefore k = -4$$

$$113. P(x) = x^4 + ax^3 + bx^2 + cx - 16$$

$$P'(x) = 4x^3 + 3ax^2 + 2bx + c$$

$$P''(x) = 12x^2 + 6ax + 2b$$

$$\text{At } x = 2,$$

$$P(2) = 16 + 8a + 4b + 2c - 16 = 0 \rightarrow 4a + 2b + c = 0 \dots (1)$$

$$P'(2) = 32 + 12a + 4b + c = 0 \rightarrow 12a + 4b + c = -32 \dots (2)$$

$$P''(2) = 48 + 12a + 2b = 0 \rightarrow 6a + b = -24 \dots (3)$$

$$(2) - (1) \quad 8a + 2b = -32$$

$$4a + b = -16 \dots (4)$$

$$(3) - (4) \quad 2a = -8 \rightarrow a = -4$$

$$\text{In (4)} \quad -16 + b = -16 \rightarrow b = 0$$

$$\text{In (1)} \quad -16 + c = 0 \rightarrow c = 16$$

$$\therefore a = -4, b = 0, c = 16$$

$$114. P(x) = x^4 + 2x^3 - 12x^2 + 14x - 5$$

$$P'(x) = 4x^3 + 6x^2 - 24x + 14$$

$$P''(x) = 12x^2 + 12x - 24$$

$$= 12(x+2)(x-1)$$

$$P(1) = 1 + 2 - 12 + 14 - 5 = 0$$

$$\therefore (x-1)^3 \text{ is a factor of } P(x)$$

$$\therefore x^4 + 2x^3 - 12x^2 + 14x - 5 \equiv (x-1)^3(x-c)$$

$$\text{Let } x = 0$$

$$-5 = -1 \times -c \rightarrow c = -5$$

$$\therefore P(x) = (x-1)^3(x+5)$$

$$\therefore \text{Roots are } 1, 1, 1, -5$$

$$115. P(x) = 2x^3 - (2p+1)x^2 + (2+q)x - 1$$

$$P'(x) = 6x^2 - 2(2p+1)x + 2+q$$

$$P'(1) = 6 - 2(2p+1) + 2+q = 0$$

$$6 - 4p - 2 + 2 + q = 0$$

$$-4p + q = -6 \dots (1)$$

$$P(1) = 2 - (2p+1) + (2+q) - 1 = 0$$

$$2 - 2p - 1 + 2 + q - 1 = 0$$

$$-2p + q = -2 \dots (2)$$

$$(1) - (2) \quad -2p = -4$$

$$p = 2$$

$$\therefore p = 2 \text{ and } q = 2$$

$$116. P(x) = x^4 - 3x^3 - 6x^2 + 28x - 24$$

$$P'(x) = 4x^3 - 9x^2 - 12x + 28$$

$$P''(x) = 12x^2 - 18x - 12$$

$$= 6(2x^2 - 3x - 2)$$

$$= 6(2x+1)(x-2)$$

$$\text{Let } P''(x) = 0$$

$$\therefore 2x+1 = 0 \text{ or } x-2 = 0$$

$$\text{For } x = 2$$

$$P(2) = 16 - 24 - 24 + 56 - 24 = 0$$

$$\therefore x = 2 \text{ is a triple zero.}$$

$$x^4 - 3x^3 - 6x^2 + 28x - 24 \equiv (x-2)^3(x-k)$$

$$\text{Let } x = 0 \quad -24 = 8k \rightarrow k = -3$$

$$\therefore P(x) = (x-2)^3(x+3)$$

$$\therefore \text{Zeros are } 2, 2, 2, -3$$

$$117. p(x) = x^3 + x^2 - 5x + 3$$

$$p'(x) = 3x^2 + 2x - 5$$

$$= (3x+5)(x-1)$$

$$x = 1 \text{ or } x = -\frac{5}{3}$$

$$p(1) = 1 + 1 - 5 + 3 = 0$$

$$\text{Roots are } 1, 1, \delta$$

$$\therefore 1 \times 1 \times \delta = -3$$

$$\therefore \delta = -3$$

$$\text{Roots are } 1, 1, -3$$

$$118. P(x) = x^4 + 4x^3 - 16x + k$$

$$P'(x) = 4x^3 + 12x^2 - 16$$

$$P''(x) = 12x^2 + 24x$$

$$= 12x(x+2) = 0$$

$$x = 0, x = -2$$

$$P(-2) = 16 - 32 + 32 + k = 0$$

$$\therefore k = -16, k \neq 0$$

$$119. P(x) = x^3 - 6x^2 + 9x + k$$

$$P'(x) = 3x^2 - 12x + 9$$

$$= 3(x-3)(x-1)$$

$$\therefore x = 3 \text{ or } x = 1$$

$$\text{If } x = 3$$

$$P(3) = 27 - 54 + 27 + k = 0$$

$$\therefore k = 0$$

$$\therefore x^3 - 6x^2 + 9x = x(x-3)^2$$

$$\text{Roots are } 0, 3, 3$$

$$\text{If } x = 1$$

$$P(1) = 1 - 6 + 9 + k = 0$$

$$\therefore k = -4$$

$$\therefore x^3 - 6x^2 + 9x - 4 = (x-1)^2(x-c)$$

$$\text{Let } x = 0$$

$$\therefore -4 = -c \rightarrow c = 4$$

$$P(x) = (x-1)^2(x-4)$$

$$\text{Roots are } 1, 1, 4$$

$$120. \quad P(x) = x^3 + 3x^2 - 24x + k$$

$$P'(x) = 3x^2 + 6x - 24$$

$$= 3(x+4)(x-2)$$

$$\text{For } x = -4$$

$$P(-4) = -64 + 48 + 96 + k = 0$$

$$\therefore k = -80$$

$$\text{For } x = 2$$

$$P(2) = 8 + 12 - 48 + k = 0$$

$$\therefore k = 28$$

$$\text{For } k = -80 \text{ and } x = -4$$

$$x^3 + 3x^2 - 24x - 80 \equiv (x+4)^2(x-c)$$

$$\text{Let } x = 0$$

$$\therefore -80 = -16c \rightarrow c = 5$$

$$P(x) = (x+4)^2(x-5)$$

$$\therefore \text{Roots are } -4, -4, 5$$

$$\text{For } k = 28 \text{ and } x = 2$$

$$x^3 + 3x^2 - 24x + 28 \equiv (x-2)^2(x-c)$$

$$\text{Let } x = 0$$

$$\therefore 28 = -4c \rightarrow c = -7$$

$$P(x) = (x-2)^2(x+7)$$

$$\therefore \text{Roots are } 2, 2, -7$$

$$121. \quad P(x) = x^3 - 7x^2 + 11x - 5$$

$$P'(x) = 3x^2 - 14x + 11$$

$$= (3x-11)(x-1)$$

$$\text{Let } P'(x) = 0$$

$$\therefore x = 3\frac{2}{3} \text{ or } x = 1$$

$$P(1) = 1 - 7 + 11 - 5 = 0$$

$$\therefore x = 1 \text{ is a double root.}$$

$$P(x) = (x-1)^2(x+k)$$

$$x^3 - 7x^2 + 11x - 5 \equiv (x-1)^2(x+k)$$

$$\text{Let } x = 0$$

$$-5 = k$$

$$P(x) = (x-1)^2(x-5)$$

$$\therefore \text{Roots are } 1, 1, 5$$

$$122. \quad y = 4x - 5 \dots\dots\dots(1)$$

$$y = x^3 - 2x^2 + 3 \dots\dots(2)$$

$$\text{Substituting (1) into (2)}$$

$$x^3 - 2x^2 + 3 = 4x - 5$$

$$\therefore P(x) = x^3 - 2x^2 - 4x + 8$$

$$P(x) = (x-2)^2(x+c)$$

$$\therefore x^3 - 2x^2 - 4x + 8 \equiv (x-2)^2(x+c)$$

$$\text{Let } x = 0$$

$$8 = 4c \rightarrow c = 2$$

$$\therefore P(x) = (x-2)^2(x+2)$$

$$\therefore x = -2, \quad y = 4 \times -2 - 5$$

$$= -13$$

$$\therefore Q(-2, -13)$$

$$123. \quad P(x) = x^3 - 3x^2 + 4$$

$$P'(x) = 3x^2 - 6x$$

$$= 3x(x-2)$$

$$\text{Let } P'(x) = 0$$

$$\therefore x = 0 \text{ or } x = 2$$

$$P(2) = 8 - 12 + 4 = 0$$

$$\therefore x = 2 \text{ is a double root.}$$

$$\therefore P(x) \text{ has a multiplicity of } 2$$

$$124. \quad \text{Let } P(x) = px^3 + qx^2 + r$$

$$P'(x) = 3px^2 + 2qx$$

$$\text{Let } P'(x) = 0$$

$$3px^2 + 2qx = 0$$

$$x(3px + 2q) = 0$$

$$\therefore x = 0 \text{ or } 3px + 2q = 0$$

$$\text{Clearly } x = 0 \text{ is not a double root.}$$

$$\therefore x = -\frac{2q}{3p} \text{ is a double root.}$$

$$P\left(-\frac{2q}{3p}\right) = p\left(-\frac{2q}{3p}\right)^3 + q\left(-\frac{2q}{3p}\right)^2 + r$$

$$= -\frac{8q^3}{27p^2} + \frac{4q^3}{9p^2} + r$$

$$= -\frac{8q^3}{27p^2} + \frac{12q^3}{27p^2} + r$$

$$= \frac{4q^3}{27p^2} + r$$

$$= \frac{4q^3 + 27p^2r}{27p^2} = 0$$

$$\therefore 27p^2r + 4q^3 = 0$$

125.

i) $y = \frac{1}{x} \dots (1)$

$(x-c)^2 + y^2 = c^2 \dots (2)$

Sub (1) into (2)

$$(x-c)^2 + \frac{1}{x^2} = c^2$$

$$x^2(x-c)^2 + 1 = c^2x^2$$

$$x^2(x^2 - 2cx + c^2) + 1 = c^2x^2$$

$$x^4 - 2cx^3 + c^2x^2 + 1 = c^2x^2$$

$$x^4 - 2cx^3 + 1 = 0$$

$\therefore$ The zeros of $P(x) = x^4 - 2cx^3 + 1$ are the x -coordinates of the points of intersection.

ii) $p(x) = x^4 - 2cx^3 + 1$

$$p'(x) = 4x^3 - 6cx^2$$

the x -intercept is a double root for only one point of intersection.

Let $p'(x) = 0 \quad 4x^3 - 6cx^2 = 0$

$$2x^2(2x - 3c) = 0$$

$$\therefore x = 0 \text{ or } x = \frac{3c}{2}$$

$P(0) = 1 \quad \therefore$ not $x = 0$

$$P\left(\frac{3c}{2}\right) = \left(\frac{3c}{2}\right)^4 - 2c\left(\frac{3c}{2}\right)^3 + 1 = 0$$

$$\frac{81c^4}{16} - \frac{27}{4}c^4 + 1 = 0$$

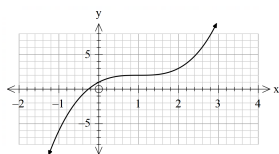
$$\frac{27}{16}c^4 = 1$$

$$c = \frac{2}{\sqrt[4]{27}} \quad c > 0$$

Polynomial Graphs

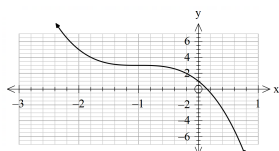
126. For $y = (x-1)^3 + 2$ the point of inflection is at (1,2)

The coefficient of x^3 is positive

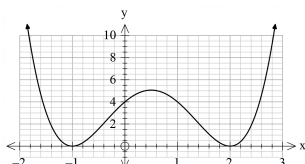


127. For $y = 3 - 2(x+1)^3$ the point of inflection is (-1,3).

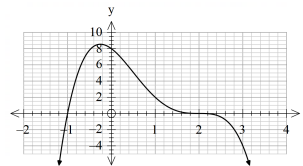
The coefficient of x^3 is negative



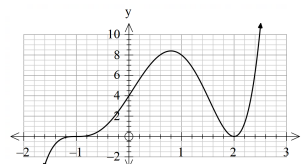
128. For $y = (x+1)^2(2-x)^2$ the x -intercepts are (-1,0) and (2,0). The coefficient of x^4 is positive



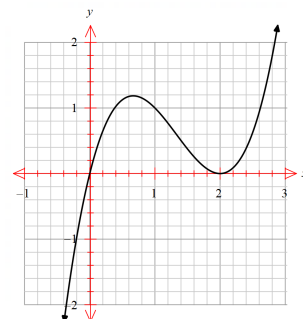
129. For $y = (x+1)(2-x)^3$ the x -intercepts are (-1,0) and (2,0) and there is a point of inflection at (2,0)



130. For $y = (x+1)^3(2-x)^2$ there is a point of inflection at (-1,0) and a turning point at (2,0). The coefficient of x^5 is positive

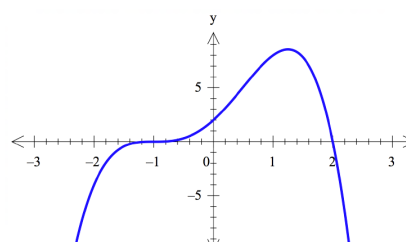


131. For $y = x(x-2)^2$ the x -intercepts are (0,0) and (2,0). The coefficient of x^3 is positive



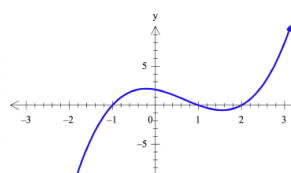
132. Let $y = 0 \rightarrow x$ intercept is (2,0), point of inflexion is (-1,0)

Let $x = 0 \rightarrow y$ intercept is (0,2)



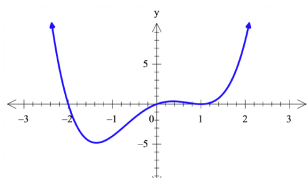
133. Let $y = 0 \rightarrow x$ intercepts are (-1,0), (1,0) and (2,0)

Let $x = 0 \rightarrow y$ intercept is (0,2)

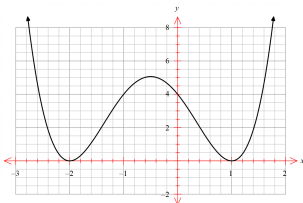


134. Let $y = 0 \rightarrow x$ intercepts are (0,0), (-2,0), point of contact is (1,0)

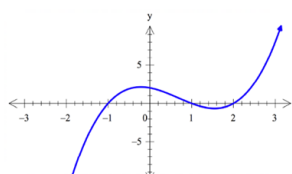
Let $x = 0 \rightarrow y$ intercept is (0,0)



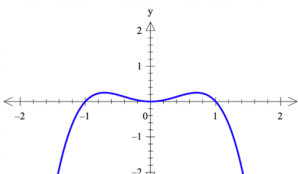
135. For $y = (x-1)^2(x+2)^2$ the x -intercepts are $(1, 0)$ are $(-2, 0)$. The coefficient of x^4 is positive



136. Let $y = 0 \rightarrow x$ intercepts are $(-1, 0)$, $(1, 0)$ and $(2, 0)$
Let $x = 0 \rightarrow y$ intercepts is $(0, 2)$



137. $P(x) = x^2(1-x^2)$
 $= x^2(1-x)(1+x) \therefore$ Zeros are $0, 1, -1$



$$\begin{array}{r} x^2 + 2x + 1 \\ x - 2 \sqrt{x^3 - 3x - 2} \\ \hline x^3 - 2x^2 \\ 2x^2 - 3x - 2 \\ 2x^2 - 4x \\ \hline x - 2 \\ x - 2 \\ \hline 0 \end{array}$$

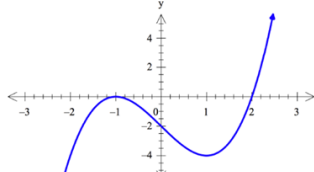
$$\therefore P(x) = (x-2)(x^2 + 2x + 1)$$

$$= (x-2)(x+1)^2$$

$$\therefore \text{Zeros are } 2, -1$$

$$y \text{ intercept is } -2$$

$$\text{By inspection } x > 2$$



139. $(x-1)^2 = x^2 - 2x + 1$

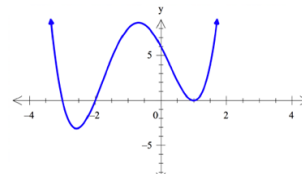
$$\begin{array}{r} x^2 + 5x + 6 \\ x^2 - 2x + 1 \sqrt{x^4 + 3x^3 - 3x^2 - 7x + 6} \\ \hline x^4 - 2x^3 + x^2 \\ \hline 5x^3 - 4x^2 - 7x + 6 \\ 5x^3 - 10x^2 + 5x \\ \hline 6x^2 - 12x + 6 \\ 6x^2 - 12x + 6 \\ \hline 0 \end{array}$$

$$\therefore P(x) = (x-1)^2(x^2 + 5x + 6)$$

$$= (x-1)^2(x+2)(x+3)$$

$$\therefore \text{Roots are } x = 1, x = -2, x = -3$$

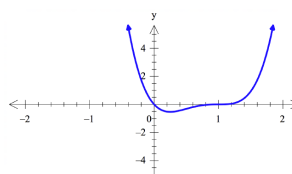
$$y \text{ intercept is } 6$$



140. Let $P(x) = 0$

$$\therefore 5x(x-1)^3 = 0$$

$$\text{Zeros are } x = 0, x = 1$$



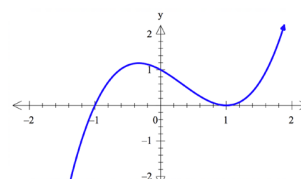
141. $P(x) = x^2(x-1) - (x-1)$

$$= (x-1)(x^2 - 1)$$

$$= (x-1)^2(x+1)$$

$$\text{Zeros are } 1, 1, -1$$

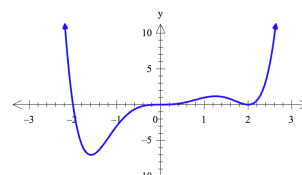
$$y \text{ intercept is } (0, 1)$$



142. Let $P(x) = 0$

$$\therefore \frac{1}{3}x^3(x-2)^2(x+2) = 0$$

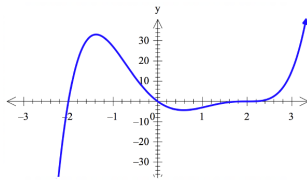
$$\therefore \text{Zeros are } 0, 2, -2$$



143. $P(x) = x(x+2)(x-2)^3$

$$\therefore x = 0, x + 2 = 0, x - 2 = 0$$

$$\therefore x = 0, x = -2, x = 2$$



144. $P(x) = -x^4 + 8x^2 - 16$

$$= -(x^4 - 8x^2 + 16)$$

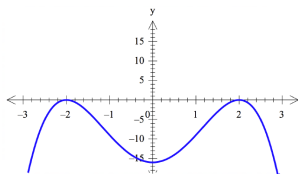
$$= -(x^2 - 4)^2$$

$$= -\{(x-2)(x+2)\}^2$$

$$= -(x-2)^2(x+2)^2$$

$\therefore$ Zeros are 2, -2

y intercept is -16



145.

(i) $P(0) = r = 0$

$$P(2) = 8 + 4p + 2q = 0 \dots (1)$$

$$P(-2) = -8 + 4p - 2q = 0 \dots (2)$$

$$(1) + (2)$$

$$8p = 0 \rightarrow p = 0$$

In (1)

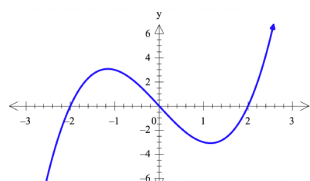
$$8 + 2q = 0 \rightarrow q = -4$$

$$\therefore p = 0, q = -4, r = 0$$

(ii) $y = x^3 - 4x$

$$= x(x^2 - 4)$$

$$= x(x-2)(x+2)$$



(iii) From the graph $-2 \leq x \leq 0, x \geq 2$

146. Given that a cubic is of the form:

$$x^3 - \sum \alpha x + \sum \alpha \beta x^2 - \alpha \beta \gamma$$

Then if $\sum \alpha > 0$ as in B then in $P(x) = ax^3 + bx^2 + cx - 6$

and $b > 0$ then B is not a solution.

$a > 0 \rightarrow$ A or B. $\therefore$ A

147. $y = a(x+2)(x+1)(x-1)^2$

From the graph $x = 0, y = 0$

$$-6 = a \times 2 \times 1 \times 1$$

$$\therefore a = -3, b = 2, c = 1, d = -1$$

148. The polynomial will be of the form

$$P(x) = (x-a)^2(x^2+x+1)$$

$$\text{Consider } g(x) = x^2 + x + 1$$

$$\Delta = (1)^2 - 4 \times 1 \times 1$$

$$= -3$$

$$\therefore \Delta < 0$$

$\therefore$ No roots.

Only 1 double root at $x = a$

$\therefore$ A or C.

$$\text{For } P(x) = (x-a)^2(x^2+x+1)$$

$$\text{For } x > a \quad P(x) > 0 \quad \therefore C$$